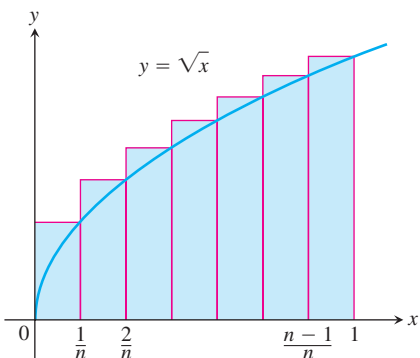


For example, let's estimate the sum of the square roots of the first n positive integers, $\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}$. The integral

$$\int_0^1 \sqrt{x} \, dx = \left. \frac{2}{3} x^{3/2} \right|_0^1 = \frac{2}{3}$$

is the limit of the upper sums

$$\begin{aligned} S_n &= \sqrt{\frac{1}{n}} \cdot \frac{1}{n} + \sqrt{\frac{2}{n}} \cdot \frac{1}{n} + \cdots + \sqrt{\frac{n}{n}} \cdot \frac{1}{n} \\ &= \frac{\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}}{n^{3/2}}. \end{aligned}$$



Therefore, when n is large, S_n will be close to $2/3$ and we will have

$$\text{Root sum} = \sqrt{1} + \sqrt{2} + \cdots + \sqrt{n} = S_n \cdot n^{3/2} \approx \frac{2}{3} n^{3/2}.$$

The following table shows how good the approximation can be.

n	Root sum	$(2/3)n^{3/2}$	Relative error
10	22.468	21.082	$1.386/22.468 \approx 6\%$
50	239.04	235.70	1.4%
100	671.46	666.67	0.7%
1000	21,097	21,082	0.07%

23. Evaluate

$$\lim_{n \rightarrow \infty} \frac{1^5 + 2^5 + 3^5 + \cdots + n^5}{n^6}$$

by showing that the limit is

$$\int_0^1 x^5 \, dx$$

and evaluating the integral.

24. See Exercise 23. Evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n^4} (1^3 + 2^3 + 3^3 + \cdots + n^3).$$

25. Let $f(x)$ be a continuous function. Express

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left[f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \cdots + f\left(\frac{n}{n}\right) \right]$$

as a definite integral.

26. Use the result of Exercise 25 to evaluate

a. $\lim_{n \rightarrow \infty} \frac{1}{n^2} (2 + 4 + 6 + \cdots + 2n),$

b. $\lim_{n \rightarrow \infty} \frac{1}{n^{16}} (1^{15} + 2^{15} + 3^{15} + \cdots + n^{15}),$

c. $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \sin \frac{3\pi}{n} + \cdots + \sin \frac{n\pi}{n} \right).$

What can be said about the following limits?

d. $\lim_{n \rightarrow \infty} \frac{1}{n^{17}} (1^{15} + 2^{15} + 3^{15} + \cdots + n^{15})$

e. $\lim_{n \rightarrow \infty} \frac{1}{n^{15}} (1^{15} + 2^{15} + 3^{15} + \cdots + n^{15})$

27. a. Show that the area A_n of an n -sided regular polygon in a circle of radius r is

$$A_n = \frac{nr^2}{2} \sin \frac{2\pi}{n}.$$

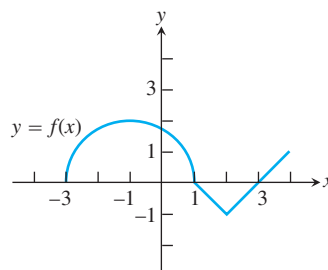
b. Find the limit of A_n as $n \rightarrow \infty$. Is this answer consistent with what you know about the area of a circle?

28. **A differential equation** Show that $y = \sin x + \int_x^\pi \cos 2t \, dt + 1$ satisfies both of the following conditions:

i. $y'' = -\sin x + 2 \sin 2x$

ii. $y = 1$ and $y' = -2$ when $x = \pi$.

29. **A function defined by an integral** The graph of a function f consists of a semicircle and two line segments as shown. Let $g(x) = \int_1^x f(t) \, dt$.



a. Find $g(1)$. b. Find $g(3)$. c. Find $g(-1)$.

d. Find all values of x on the open interval $(-3, 4)$ at which g has a relative maximum.

e. Write an equation for the line tangent to the graph of g at $x = -1$.

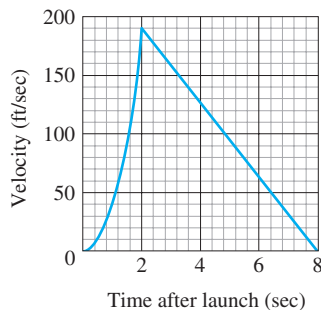
f. Find the x -coordinate of each point of inflection of the graph of g on the open interval $(-3, 4)$.

g. Find the range of g .

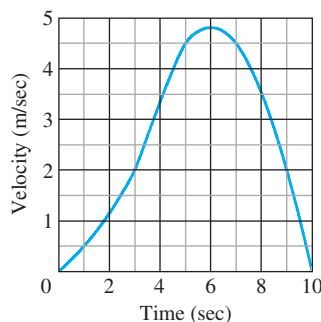
Chapter 5 Practice Exercises

Finite Sums and Estimates

1. The accompanying figure shows the graph of the velocity (ft/sec) of a model rocket for the first 8 sec after launch. The rocket accelerated straight up for the first 2 sec and then coasted to reach its maximum height at $t = 8$ sec.



- a. Assuming that the rocket was launched from ground level, about how high did it go? (This is the rocket in Section 3.3, Exercise 17, but you do not need to do Exercise 17 to do the exercise here.)
- b. Sketch a graph of the rocket's height aboveground as a function of time for $0 \leq t \leq 8$.
2. a. The accompanying figure shows the velocity (m/sec) of a body moving along the s -axis during the time interval from $t = 0$ to $t = 10$ sec. About how far did the body travel during those 10 sec?
- b. Sketch a graph of s as a function of t for $0 \leq t \leq 10$ assuming $s(0) = 0$.



3. Suppose that $\sum_{k=1}^{10} a_k = -2$ and $\sum_{k=1}^{10} b_k = 25$. Find the value of

a. $\sum_{k=1}^{10} \frac{a_k}{4}$ b. $\sum_{k=1}^{10} (b_k - 3a_k)$

c. $\sum_{k=1}^{10} (a_k + b_k - 1)$ d. $\sum_{k=1}^{10} \left(\frac{5}{2} - b_k \right)$

4. Suppose that $\sum_{k=1}^{20} a_k = 0$ and $\sum_{k=1}^{20} b_k = 7$. Find the values of

a. $\sum_{k=1}^{20} 3a_k$ b. $\sum_{k=1}^{20} (a_k + b_k)$

c. $\sum_{k=1}^{20} \left(\frac{1}{2} - \frac{2b_k}{7} \right)$ d. $\sum_{k=1}^{20} (a_k - 2)$

Definite Integrals

In Exercises 5–8, express each limit as a definite integral. Then evaluate the integral to find the value of the limit. In each case, P is a partition of the given interval and the numbers c_k are chosen from the subintervals of P .

5. $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (2c_k - 1)^{-1/2} \Delta x_k$, where P is a partition of $[1, 5]$
6. $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n c_k (c_k^2 - 1)^{1/3} \Delta x_k$, where P is a partition of $[1, 3]$
7. $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \left(\cos \left(\frac{c_k}{2} \right) \right) \Delta x_k$, where P is a partition of $[-\pi, 0]$
8. $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (\sin c_k)(\cos c_k) \Delta x_k$, where P is a partition of $[0, \pi/2]$
9. If $\int_{-2}^2 3f(x) dx = 12$, $\int_{-2}^5 f(x) dx = 6$, and $\int_{-2}^5 g(x) dx = 2$, find the values of the following.

a. $\int_{-2}^2 f(x) dx$ b. $\int_2^5 f(x) dx$

c. $\int_5^{-2} g(x) dx$ d. $\int_{-2}^5 (-\pi g(x)) dx$

e. $\int_{-2}^5 \left(\frac{f(x) + g(x)}{5} \right) dx$

10. If $\int_0^2 f(x) dx = \pi$, $\int_0^2 7g(x) dx = 7$, and $\int_0^1 g(x) dx = 2$, find the values of the following.

a. $\int_0^2 g(x) dx$ b. $\int_1^2 g(x) dx$

c. $\int_2^0 f(x) dx$ d. $\int_0^2 \sqrt{2} f(x) dx$

e. $\int_0^2 (g(x) - 3f(x)) dx$

Area

In Exercise 11–14, find the total area of the region between the graph of f and the x -axis.

11. $f(x) = x^2 - 4x + 3$, $0 \leq x \leq 3$
12. $f(x) = 1 - (x^2/4)$, $-2 \leq x \leq 3$

13. $f(x) = 5 - 5x^{2/3}, \quad -1 \leq x \leq 8$

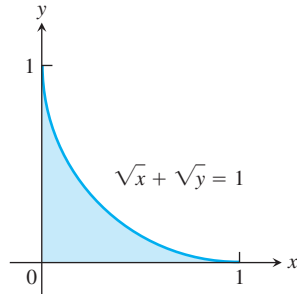
14. $f(x) = 1 - \sqrt{x}, \quad 0 \leq x \leq 4$

Find the areas of the regions enclosed by the curves and lines in Exercises 15–26.

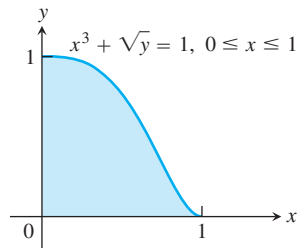
15. $y = x, \quad y = 1/x^2, \quad x = 2$

16. $y = x, \quad y = 1/\sqrt{x}, \quad x = 2$

17. $\sqrt{x} + \sqrt{y} = 1, \quad x = 0, \quad y = 0$



18. $x^3 + \sqrt{y} = 1, \quad x = 0, \quad y = 0, \quad \text{for } 0 \leq x \leq 1$



19. $x = 2y^2, \quad x = 0, \quad y = 3$ 20. $x = 4 - y^2, \quad x = 0$

21. $y^2 = 4x, \quad y = 4x - 2$

22. $y^2 = 4x + 4, \quad y = 4x - 16$

23. $y = \sin x, \quad y = x, \quad 0 \leq x \leq \pi/4$

24. $y = |\sin x|, \quad y = 1, \quad -\pi/2 \leq x \leq \pi/2$

25. $y = 2 \sin x, \quad y = \sin 2x, \quad 0 \leq x \leq \pi$

26. $y = 8 \cos x, \quad y = \sec^2 x, \quad -\pi/3 \leq x \leq \pi/3$

27. Find the area of the “triangular” region bounded on the left by $x + y = 2$, on the right by $y = x^2$, and above by $y = 2$.

28. Find the area of the “triangular” region bounded on the left by $y = \sqrt{x}$, on the right by $y = 6 - x$, and below by $y = 1$.

29. Find the extreme values of $f(x) = x^3 - 3x^2$ and find the area of the region enclosed by the graph of f and the x -axis.

30. Find the area of the region cut from the first quadrant by the curve $x^{1/2} + y^{1/2} = a^{1/2}$.

31. Find the total area of the region enclosed by the curve $x = y^{2/3}$ and the lines $x = y$ and $y = -1$.

32. Find the total area of the region between the curves $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq 3\pi/2$.

Initial Value Problems

33. Show that $y = x^2 + \int_1^x \frac{1}{t} dt$ solves the initial value problem

$$\frac{d^2 y}{dx^2} = 2 - \frac{1}{x^2}; \quad y'(1) = 3, \quad y(1) = 1.$$

34. Show that $y = \int_0^x (1 + 2\sqrt{\sec t}) dt$ solves the initial value problem

$$\frac{d^2 y}{dx^2} = \sqrt{\sec x} \tan x; \quad y'(0) = 3, \quad y(0) = 0.$$

Express the solutions of the initial value problems in Exercises 35 and 36 in terms of integrals.

35. $\frac{dy}{dx} = \frac{\sin x}{x}, \quad y(5) = -3$

36. $\frac{dy}{dx} = \sqrt{2 - \sin^2 x}, \quad y(-1) = 2$

Evaluating Indefinite Integrals

Evaluate the integrals in Exercises 37–44.

37. $\int 2(\cos x)^{-1/2} \sin x dx$ 38. $\int (\tan x)^{-3/2} \sec^2 x dx$

39. $\int (2\theta + 1 + 2 \cos(2\theta + 1)) d\theta$

40. $\int \left(\frac{1}{\sqrt{2\theta - \pi}} + 2 \sec^2(2\theta - \pi) \right) d\theta$

41. $\int \left(t - \frac{2}{t} \right) \left(t + \frac{2}{t} \right) dt$ 42. $\int \frac{(t+1)^2 - 1}{t^4} dt$

43. $\int \sqrt{t} \sin(2t^{3/2}) dt$ 44. $\int \sec \theta \tan \theta \sqrt{1 + \sec \theta} d\theta$

Evaluating Definite Integrals

Evaluate the integrals in Exercises 45–70.

45. $\int_{-1}^1 (3x^2 - 4x + 7) dx$ 46. $\int_0^1 (8s^3 - 12s^2 + 5) ds$

47. $\int_1^2 \frac{4}{v^2} dv$ 48. $\int_1^{27} x^{-4/3} dx$

49. $\int_1^4 \frac{dt}{t\sqrt{t}}$ 50. $\int_1^4 \frac{(1 + \sqrt{u})^{1/2}}{\sqrt{u}} du$

51. $\int_0^1 \frac{36 dx}{(2x + 1)^3}$ 52. $\int_0^1 \frac{dr}{\sqrt[3]{(7 - 5r)^2}}$

53. $\int_{1/8}^1 x^{-1/3} (1 - x^{2/3})^{3/2} dx$ 54. $\int_0^{1/2} x^3 (1 + 9x^4)^{-3/2} dx$

55. $\int_0^\pi \sin^2 5r dr$ 56. $\int_0^{\pi/4} \cos^2 \left(4t - \frac{\pi}{4} \right) dt$

57. $\int_0^{\pi/3} \sec^2 \theta \, d\theta$ 58. $\int_{\pi/4}^{3\pi/4} \csc^2 x \, dx$
59. $\int_{\pi}^{3\pi} \cot^2 \frac{x}{6} \, dx$ 60. $\int_0^{\pi} \tan^2 \frac{\theta}{3} \, d\theta$
61. $\int_{-\pi/3}^0 \sec x \tan x \, dx$ 62. $\int_{\pi/4}^{3\pi/4} \csc z \cot z \, dz$
63. $\int_0^{\pi/2} 5(\sin x)^{3/2} \cos x \, dx$ 64. $\int_{-1}^1 2x \sin(1 - x^2) \, dx$
65. $\int_{-\pi/2}^{\pi/2} 15 \sin^4 3x \cos 3x \, dx$ 66. $\int_0^{2\pi/3} \cos^{-4} \left(\frac{x}{2} \right) \sin \left(\frac{x}{2} \right) \, dx$
67. $\int_0^{\pi/2} \frac{3 \sin x \cos x}{\sqrt{1 + 3 \sin^2 x}} \, dx$ 68. $\int_0^{\pi/4} \frac{\sec^2 x}{(1 + 7 \tan x)^{2/3}} \, dx$
69. $\int_0^{\pi/3} \frac{\tan \theta}{\sqrt{2 \sec \theta}} \, d\theta$ 70. $\int_{\pi^2/36}^{\pi^2/4} \frac{\cos \sqrt{t}}{\sqrt{t \sin \sqrt{t}}} \, dt$

Average Values

71. Find the average value of $f(x) = mx + b$
- over $[-1, 1]$
 - over $[-k, k]$
72. Find the average value of
- $y = \sqrt{3x}$ over $[0, 3]$
 - $y = \sqrt{ax}$ over $[0, a]$
73. Let f be a function that is differentiable on $[a, b]$. In Chapter 2 we defined the average rate of change of f over $[a, b]$ to be

$$\frac{f(b) - f(a)}{b - a}$$

and the instantaneous rate of change of f at x to be $f'(x)$. In this chapter we defined the average value of a function. For the new definition of average to be consistent with the old one, we should have

$$\frac{f(b) - f(a)}{b - a} = \text{average value of } f' \text{ on } [a, b].$$

Is this the case? Give reasons for your answer.

74. Is it true that the average value of an integrable function over an interval of length 2 is half the function's integral over the interval? Give reasons for your answer.

- T** 75. Compute the average value of the temperature function

$$f(x) = 37 \sin \left(\frac{2\pi}{365} (x - 101) \right) + 25$$

for a 365-day year. This is one way to estimate the annual mean air temperature in Fairbanks, Alaska. The National Weather Service's official figure, a numerical average of the daily normal mean air temperatures for the year, is 25.7°F , which is slightly higher than the average value of $f(x)$. Figure 3.33 shows why.

- T** 76. **Specific heat of a gas** Specific heat C_v is the amount of heat required to raise the temperature of a given mass of gas with con-

stant volume by 1°C , measured in units of cal/deg-mole (calories per degree gram molecule). The specific heat of oxygen depends on its temperature T and satisfies the formula

$$C_v = 8.27 + 10^{-5} (26T - 1.87T^2).$$

Find the average value of C_v for $20^\circ \leq T \leq 675^\circ\text{C}$ and the temperature at which it is attained.

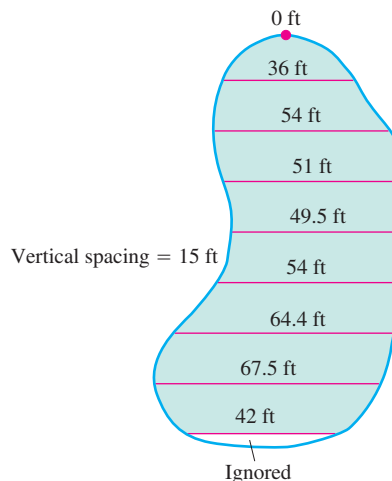
Differentiating Integrals

In Exercises 77–80, find dy/dx .

77. $y = \int_2^x \sqrt{2 + \cos^3 t} \, dt$ 78. $y = \int_2^{7x^2} \sqrt{2 + \cos^3 t} \, dt$
79. $y = \int_x^1 \frac{6}{3 + t^4} \, dt$ 80. $y = \int_{\sec x}^2 \frac{1}{t^2 + 1} \, dt$

Theory and Examples

81. Is it true that every function $y = f(x)$ that is differentiable on $[a, b]$ is itself the derivative of some function on $[a, b]$? Give reasons for your answer.
82. Suppose that $F(x)$ is an antiderivative of $f(x) = \sqrt{1 + x^4}$. Express $\int_0^1 \sqrt{1 + x^4} \, dx$ in terms of F and give a reason for your answer.
83. Find dy/dx if $y = \int_x^1 \sqrt{1 + t^2} \, dt$. Explain the main steps in your calculation.
84. Find dy/dx if $y = \int_{\cos x}^0 (1/(1 - t^2)) \, dt$. Explain the main steps in your calculation.
85. **A new parking lot** To meet the demand for parking, your town has allocated the area shown here. As the town engineer, you have been asked by the town council to find out if the lot can be built for \$10,000. The cost to clear the land will be \$0.10 a square foot, and the lot will cost \$2.00 a square foot to pave. Can the job be done for \$10,000? Use a lower sum estimate to see. (Answers may vary slightly, depending on the estimate used.)



86. Skydivers A and B are in a helicopter hovering at 6400 ft. Skydiver A jumps and descends for 4 sec before opening her parachute. The helicopter then climbs to 7000 ft and hovers there. Forty-five seconds after A leaves the aircraft, B jumps and descends for 13 sec before opening his parachute. Both skydivers descend at 16 ft/sec with parachutes open. Assume that the skydivers fall freely (no effective air resistance) before their parachutes open.
- At what altitude does A's parachute open?
 - At what altitude does B's parachute open?
 - Which skydiver lands first?

Average Daily Inventory

Average value is used in economics to study such things as average daily inventory. If $I(t)$ is the number of radios, tires, shoes, or whatever product a firm has on hand on day t (we call I an **inventory function**), the average value of I over a time period $[0, T]$ is called the firm's average daily inventory for the period.

$$\text{Average daily inventory} = \text{av}(I) = \frac{1}{T} \int_0^T I(t) \, dt.$$

If h is the dollar cost of holding one item per day, the product $\text{av}(I) \cdot h$ is the **average daily holding cost** for the period.

87. As a wholesaler, Tracey Burr Distributors receives a shipment of 1200 cases of chocolate bars every 30 days. TBD sells the chocolate to retailers at a steady rate, and t days after a shipment arrives, its inventory of cases on hand is $I(t) = 1200 - 40t$, $0 \leq t \leq 30$. What is TBD's average daily inventory for the 30-day period? What is its average daily holding cost if the cost of holding one case is 3¢ a day?
88. Rich Wholesale Foods, a manufacturer of cookies, stores its cases of cookies in an air-conditioned warehouse for shipment every 14 days. Rich tries to keep 600 cases on reserve to meet occasional peaks in demand, so a typical 14-day inventory function is $I(t) = 600 + 600t$, $0 \leq t \leq 14$. The daily holding cost for each case is 4¢ per day. Find Rich's average daily inventory and average daily holding cost.
89. Solon Container receives 450 drums of plastic pellets every 30 days. The inventory function (drums on hand as a function of days) is $I(t) = 450 - t^2/2$. Find the average daily inventory. If the holding cost for one drum is 2¢ per day, find the average daily holding cost.
90. Mitchell Mailorder receives a shipment of 600 cases of athletic socks every 60 days. The number of cases on hand t days after the shipment arrives is $I(t) = 600 - 20\sqrt{15t}$. Find the average daily inventory. If the holding cost for one case is 1/2¢ per day, find the average daily holding cost.

Chapter 5 Questions to Guide Your Review

1. How can you sometimes estimate quantities like distance traveled, area, and average value with finite sums? Why might you want to do so?
2. What is sigma notation? What advantage does it offer? Give examples.
3. What is a Riemann sum? Why might you want to consider such a sum?
4. What is the norm of a partition of a closed interval?
5. What is the definite integral of a function f over a closed interval $[a, b]$? When can you be sure it exists?
6. What is the relation between definite integrals and area? Describe some other interpretations of definite integrals.
7. What is the average value of an integrable function over a closed interval? Must the function assume its average value? Explain.
8. Describe the rules for working with definite integrals (Table 5.3). Give examples.
9. What is the Fundamental Theorem of Calculus? Why is it so important? Illustrate each part of the theorem with an example.
10. How does the Fundamental Theorem provide a solution to the initial value problem $dy/dx = f(x)$, $y(x_0) = y_0$, when f is continuous?
11. How is integration by substitution related to the Chain Rule?
12. How can you sometimes evaluate indefinite integrals by substitution? Give examples.
13. How does the method of substitution work for definite integrals? Give examples.
14. How do you define and calculate the area of the region between the graphs of two continuous functions? Give an example.

Chapter 5 Technology Application Projects

Mathematica/Maple Module

Using Riemann Sums to Estimate Areas, Volumes, and Lengths of Curves

Visualize and approximate areas and volumes in Part I.

Mathematica/Maple Module

Riemann Sums, Definite Integrals, and the Fundamental Theorem of Calculus

Parts I, II, and III develop Riemann sums and definite integrals. Part IV continues the development of the Riemann sum and definite integral using the Fundamental Theorem to solve problems previously investigated.

Mathematica/Maple Module

Rain Catchers, Elevators, and Rockets

Part I illustrates that the area under a curve is the same as the area of an appropriate rectangle for examples taken from the chapter. You will compute the amount of water accumulating in basins of different shapes as the basin is filled and drained.

Mathematica/Maple Module

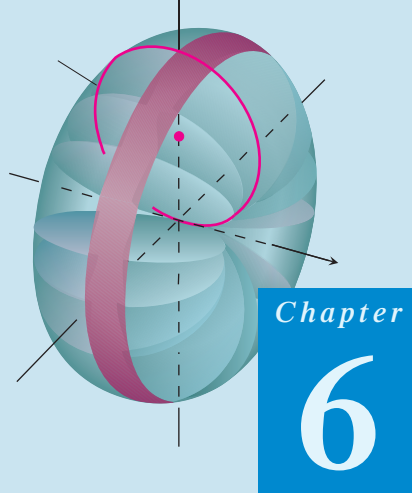
Motion Along a Straight Line, Part II

You will observe the shape of a graph through dramatic animated visualizations of the derivative relations among the position, velocity, and acceleration. Figures in the text can be animated using this software.

Mathematica/Maple Module

Bending of Beams

Study bent shapes of beams, determine their maximum deflections, concavity and inflection points, and interpret the results in terms of a beam's compression and tension.



APPLICATIONS OF DEFINITE INTEGRALS

OVERVIEW In Chapter 5 we discovered the connection between Riemann sums

$$S_P = \sum_{k=1}^n f(c_k) \Delta x_k$$

associated with a partition P of the finite closed interval $[a, b]$ and the process of integration. We found that for a continuous function f on $[a, b]$, the limit of S_P as the norm of the partition $\|P\|$ approaches zero is the number

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f . We applied this to the problems of computing the area between the x -axis and the graph of $y = f(x)$ for $a \leq x \leq b$, and to finding the area between two curves.

In this chapter we extend the applications to finding volumes, lengths of plane curves, centers of mass, areas of surfaces of revolution, work, and fluid forces against planar walls. We define all these as limits of Riemann sums of continuous functions on closed intervals—that is, as definite integrals which can be evaluated using the Fundamental Theorem of Calculus.

6.1

Volumes by Slicing and Rotation About an Axis

In this section we define volumes of solids whose cross-sections are plane regions. A **cross-section** of a solid S is the plane region formed by intersecting S with a plane (Figure 6.1).

Suppose we want to find the volume of a solid S like the one in Figure 6.1. We begin by extending the definition of a cylinder from classical geometry to cylindrical solids with arbitrary bases (Figure 6.2). If the cylindrical solid has a known base area A and height h , then the volume of the cylindrical solid is

$$\text{Volume} = \text{area} \times \text{height} = A \cdot h.$$

This equation forms the basis for defining the volumes of many solids that are not cylindrical by the *method of slicing*.

If the cross-section of the solid S at each point x in the interval $[a, b]$ is a region $R(x)$ of area $A(x)$, and A is a continuous function of x , we can define and calculate the volume of the solid S as a definite integral in the following way.

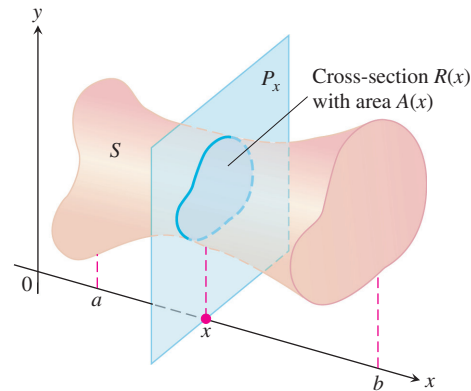


FIGURE 6.1 A cross-section of the solid S formed by intersecting S with a plane P_x perpendicular to the x -axis through the point x in the interval $[a, b]$.

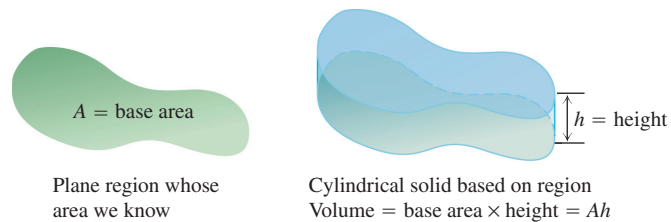


FIGURE 6.2 The volume of a cylindrical solid is always defined to be its base area times its height.

We partition $[a, b]$ into subintervals of width (length) Δx_k and slice the solid, as we would a loaf of bread, by planes perpendicular to the x -axis at the partition points $a = x_0 < x_1 < \cdots < x_n = b$. The planes P_{x_k} , perpendicular to the x -axis at the partition points, slice S into thin “slabs” (like thin slices of a loaf of bread). A typical slab is shown in Figure 6.3. We approximate the slab between the plane at x_{k-1} and the plane at x_k by a cylindrical solid with base area $A(x_k)$ and height $\Delta x_k = x_k - x_{k-1}$ (Figure 6.4). The volume V_k of this cylindrical solid is $A(x_k) \cdot \Delta x_k$, which is approximately the same volume as that of the slab:

$$\text{Volume of the } k\text{th slab} \approx V_k = A(x_k) \Delta x_k.$$

The volume V of the entire solid S is therefore approximated by the sum of these cylindrical volumes,

$$V \approx \sum_{k=1}^n V_k = \sum_{k=1}^n A(x_k) \Delta x_k.$$

This is a Riemann sum for the function $A(x)$ on $[a, b]$. We expect the approximations from these sums to improve as the norm of the partition of $[a, b]$ goes to zero, so we define their limiting definite integral to be the volume of the solid S .

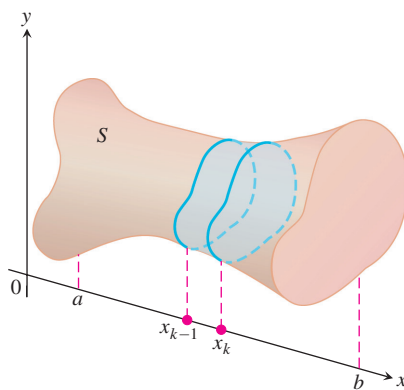


FIGURE 6.3 A typical thin slab in the solid S .

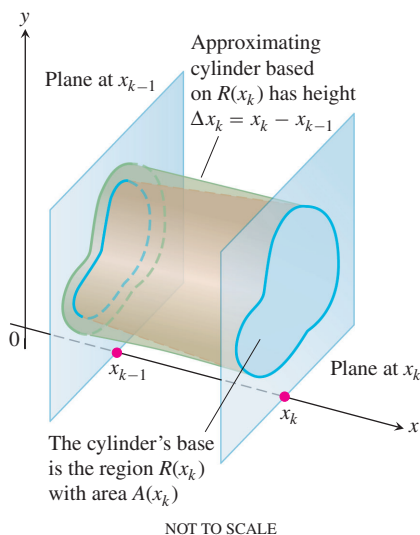


FIGURE 6.4 The solid thin slab in Figure 6.3 is approximated by the cylindrical solid with base $R(x_k)$ having area $A(x_k)$ and height $\Delta x_k = x_k - x_{k-1}$.

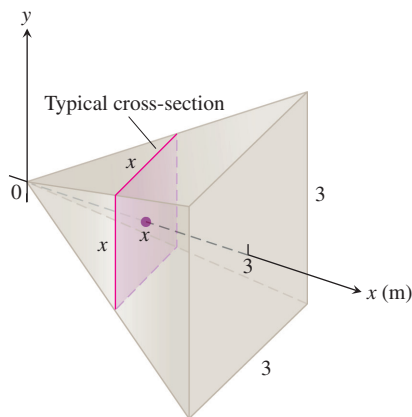


FIGURE 6.5 The cross-sections of the pyramid in Example 1 are squares.

DEFINITION Volume

The **volume** of a solid of known integrable cross-sectional area $A(x)$ from $x = a$ to $x = b$ is the integral of A from a to b ,

$$V = \int_a^b A(x) \, dx.$$

This definition applies whenever $A(x)$ is continuous, or more generally, when it is integrable. To apply the formula in the definition to calculate the volume of a solid, take the following steps:

Calculating the Volume of a Solid

1. Sketch the solid and a typical cross-section.
2. Find a formula for $A(x)$, the area of a typical cross-section.
3. Find the limits of integration.
4. Integrate $A(x)$ using the Fundamental Theorem.

EXAMPLE 1 Volume of a Pyramid

A pyramid 3 m high has a square base that is 3 m on a side. The cross-section of the pyramid perpendicular to the altitude x m down from the vertex is a square x m on a side. Find the volume of the pyramid.

Solution

1. *A sketch.* We draw the pyramid with its altitude along the x -axis and its vertex at the origin and include a typical cross-section (Figure 6.5).
2. *A formula for $A(x)$.* The cross-section at x is a square x meters on a side, so its area is
$$A(x) = x^2.$$
3. *The limits of integration.* The squares lie on the planes from $x = 0$ to $x = 3$.
4. *Integrate to find the volume.*

$$V = \int_0^3 A(x) \, dx = \int_0^3 x^2 \, dx = \left. \frac{x^3}{3} \right|_0^3 = 9 \, \text{m}^3$$

EXAMPLE 2 Cavalieri's Principle

Cavalieri's principle says that solids with equal altitudes and identical cross-sectional areas at each height have the same volume (Figure 6.6). This follows immediately from the definition of volume, because the cross-sectional area function $A(x)$ and the interval $[a, b]$ are the same for both solids.

HISTORICAL BIOGRAPHY

Bonaventura Cavalieri
(1598–1647)

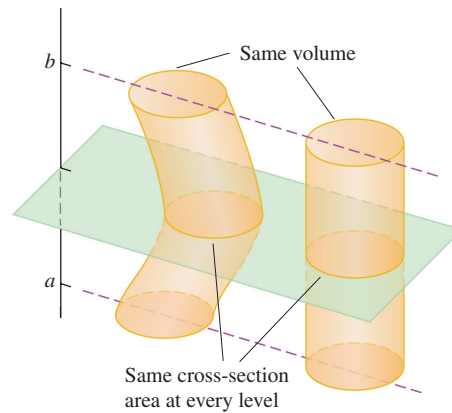


FIGURE 6.6 Cavalieri's Principle: These solids have the same volume, which can be illustrated with stacks of coins (Example 2).

EXAMPLE 3 Volume of a Wedge

A curved wedge is cut from a cylinder of radius 3 by two planes. One plane is perpendicular to the axis of the cylinder. The second plane crosses the first plane at a 45° angle at the center of the cylinder. Find the volume of the wedge.

Solution We draw the wedge and sketch a typical cross-section perpendicular to the x -axis (Figure 6.7). The cross-section at x is a rectangle of area

$$\begin{aligned} A(x) &= (\text{height})(\text{width}) = (x)(2\sqrt{9 - x^2}) \\ &= 2x\sqrt{9 - x^2}. \end{aligned}$$

The rectangles run from $x = 0$ to $x = 3$, so we have

$$\begin{aligned} V &= \int_a^b A(x) \, dx = \int_0^3 2x\sqrt{9 - x^2} \, dx \\ &= -\frac{2}{3} (9 - x^2)^{3/2} \Big|_0^3 \\ &= 0 + \frac{2}{3} (9)^{3/2} \\ &= 18. \end{aligned}$$

Let $u = 9 - x^2$,
 $du = -2x \, dx$, integrate,
and substitute back.

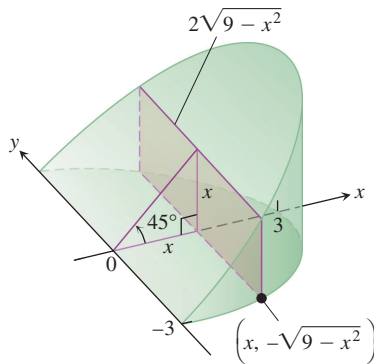


FIGURE 6.7 The wedge of Example 3, sliced perpendicular to the x -axis. The cross-sections are rectangles.

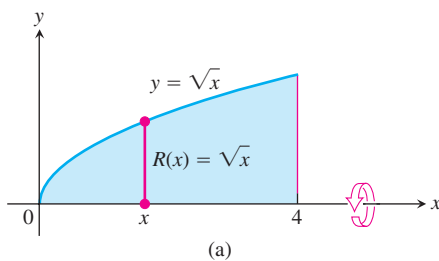
Solids of Revolution: The Disk Method

The solid generated by rotating a plane region about an axis in its plane is called a **solid of revolution**. To find the volume of a solid like the one shown in Figure 6.8, we need only observe that the cross-sectional area $A(x)$ is the area of a disk of radius $R(x)$, the distance of the planar region's boundary from the axis of revolution. The area is then

$$A(x) = \pi(\text{radius})^2 = \pi[R(x)]^2.$$

So the definition of volume gives

$$V = \int_a^b A(x) \, dx = \int_a^b \pi[R(x)]^2 \, dx.$$



This method for calculating the volume of a solid of revolution is often called the **disk method** because a cross-section is a circular disk of radius $R(x)$.

EXAMPLE 4 A Solid of Revolution (Rotation About the x -Axis)

The region between the curve $y = \sqrt{x}$, $0 \leq x \leq 4$, and the x -axis is revolved about the x -axis to generate a solid. Find its volume.

Solution We draw figures showing the region, a typical radius, and the generated solid (Figure 6.8). The volume is

$$\begin{aligned} V &= \int_a^b \pi [R(x)]^2 dx \\ &= \int_0^4 \pi [\sqrt{x}]^2 dx && R(x) = \sqrt{x} \\ &= \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = \pi \frac{(4)^2}{2} = 8\pi. \end{aligned}$$

EXAMPLE 5 Volume of a Sphere

The circle

$$x^2 + y^2 = a^2$$

is rotated about the x -axis to generate a sphere. Find its volume.

Solution We imagine the sphere cut into thin slices by planes perpendicular to the x -axis (Figure 6.9). The cross-sectional area at a typical point x between $-a$ and a is

$$A(x) = \pi y^2 = \pi(a^2 - x^2).$$

Therefore, the volume is

$$V = \int_{-a}^a A(x) dx = \int_{-a}^a \pi(a^2 - x^2) dx = \pi \left[a^2x - \frac{x^3}{3} \right]_{-a}^a = \frac{4}{3} \pi a^3.$$

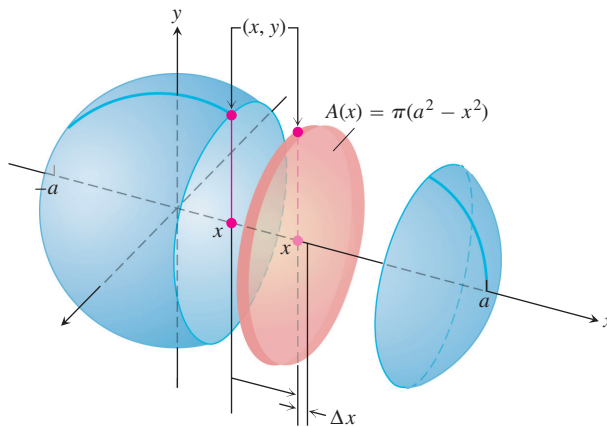


FIGURE 6.9 The sphere generated by rotating the circle $x^2 + y^2 = a^2$ about the x -axis. The radius is $R(x) = y = \sqrt{a^2 - x^2}$ (Example 5).

The axis of revolution in the next example is not the x -axis, but the rule for calculating the volume is the same: Integrate $\pi(\text{radius})^2$ between appropriate limits.

EXAMPLE 6 A Solid of Revolution (Rotation About the Line $y = 1$)

Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 1$, $x = 4$ about the line $y = 1$.

Solution We draw figures showing the region, a typical radius, and the generated solid (Figure 6.10). The volume is

$$\begin{aligned} V &= \int_1^4 \pi[R(x)]^2 dx \\ &= \int_1^4 \pi[\sqrt{x} - 1]^2 dx \\ &= \pi \int_1^4 [x - 2\sqrt{x} + 1] dx \\ &= \pi \left[\frac{x^2}{2} - 2 \cdot \frac{2}{3} x^{3/2} + x \right]_1^4 = \frac{7\pi}{6}. \end{aligned}$$

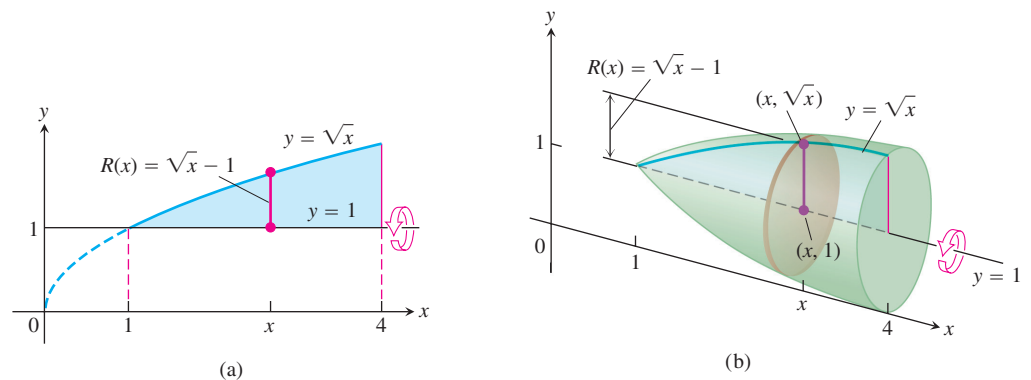


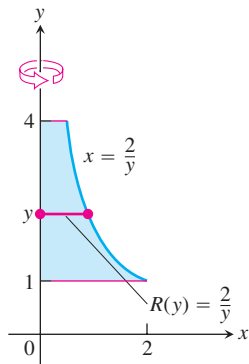
FIGURE 6.10 The region (a) and solid of revolution (b) in Example 6.

To find the volume of a solid generated by revolving a region between the y -axis and a curve $x = R(y)$, $c \leq y \leq d$, about the y -axis, we use the same method with x replaced by y . In this case, the circular cross-section is

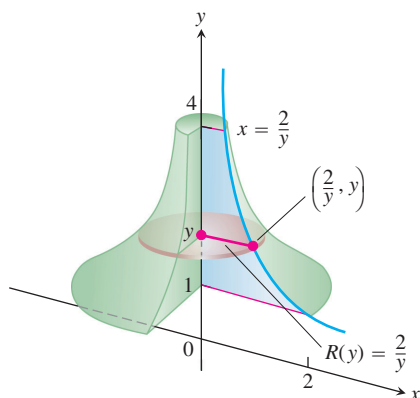
$$A(y) = \pi[\text{radius}]^2 = \pi[R(y)]^2.$$

EXAMPLE 7 Rotation About the y -Axis

Find the volume of the solid generated by revolving the region between the y -axis and the curve $x = 2/y$, $1 \leq y \leq 4$, about the y -axis.



(a)


FIGURE 6.11 The region (a) and part of the solid of revolution (b) in Example 7.

Solution We draw figures showing the region, a typical radius, and the generated solid (Figure 6.11). The volume is

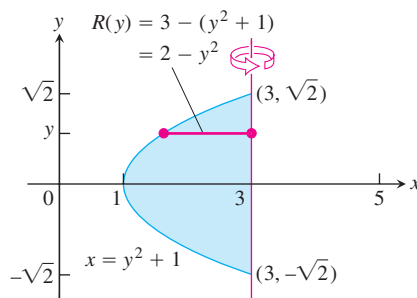
$$\begin{aligned} V &= \int_1^4 \pi [R(y)]^2 dy \\ &= \int_1^4 \pi \left(\frac{2}{y}\right)^2 dy \\ &= \pi \int_1^4 \frac{4}{y^2} dy = 4\pi \left[-\frac{1}{y}\right]_1^4 = 4\pi \left[\frac{3}{4}\right] \\ &= 3\pi. \end{aligned}$$

EXAMPLE 8 Rotation About a Vertical Axis

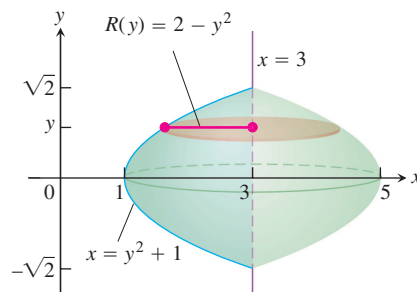
Find the volume of the solid generated by revolving the region between the parabola $x = y^2 + 1$ and the line $x = 3$ about the line $x = 3$.

Solution We draw figures showing the region, a typical radius, and the generated solid (Figure 6.12). Note that the cross-sections are perpendicular to the line $x = 3$. The volume is

$$\begin{aligned} V &= \int_{-\sqrt{2}}^{\sqrt{2}} \pi [R(y)]^2 dy \\ &= \int_{-\sqrt{2}}^{\sqrt{2}} \pi [2 - y^2]^2 dy \quad \begin{array}{l} R(y) = 3 - (y^2 + 1) \\ \quad = 2 - y^2 \end{array} \\ &= \pi \int_{-\sqrt{2}}^{\sqrt{2}} [4 - 4y^2 + y^4] dy \\ &= \pi \left[4y - \frac{4}{3}y^3 + \frac{y^5}{5} \right]_{-\sqrt{2}}^{\sqrt{2}} \\ &= \frac{64\pi\sqrt{2}}{15}. \end{aligned}$$



(a)



(b)

FIGURE 6.12 The region (a) and solid of revolution (b) in Example 8.

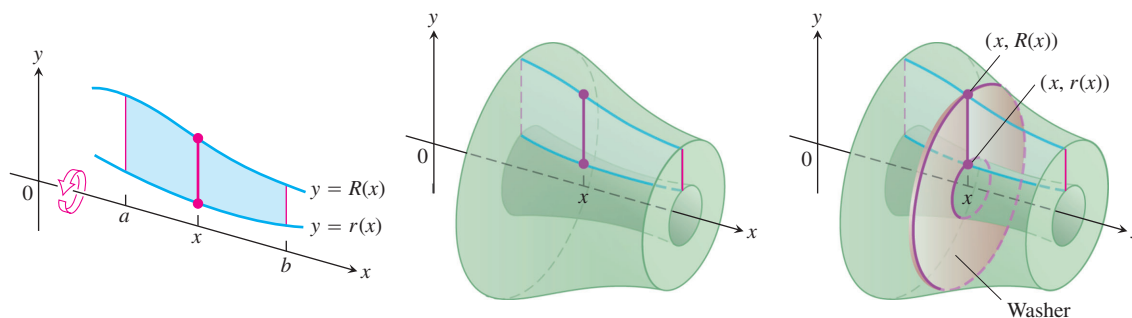


FIGURE 6.13 The cross-sections of the solid of revolution generated here are washers, not disks, so the integral $\int_a^b A(x) dx$ leads to a slightly different formula.

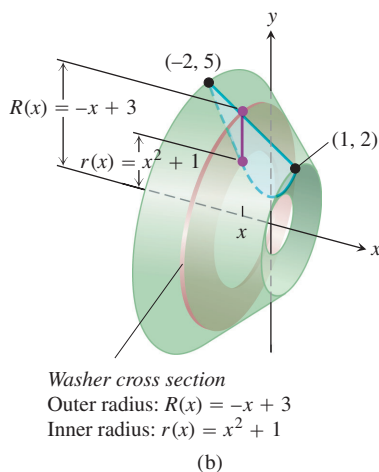
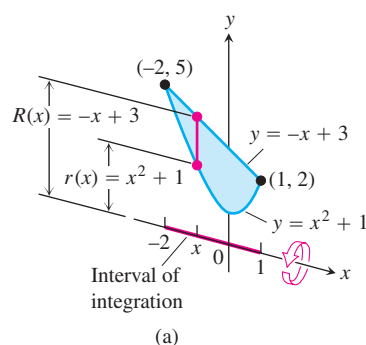


FIGURE 6.14 (a) The region in Example 9 spanned by a line segment perpendicular to the axis of revolution. (b) When the region is revolved about the x -axis, the line segment generates a washer.

Solids of Revolution: The Washer Method

If the region we revolve to generate a solid does not border on or cross the axis of revolution, the solid has a hole in it (Figure 6.13). The cross-sections perpendicular to the axis of revolution are washers (the purplish circular surface in Figure 6.13) instead of disks. The dimensions of a typical washer are

$$\text{Outer radius: } R(x)$$

$$\text{Inner radius: } r(x)$$

The washer's area is

$$A(x) = \pi[R(x)]^2 - \pi[r(x)]^2 = \pi([R(x)]^2 - [r(x)]^2).$$

Consequently, the definition of volume gives

$$V = \int_a^b A(x) dx = \int_a^b \pi([R(x)]^2 - [r(x)]^2) dx.$$

This method for calculating the volume of a solid of revolution is called the **washer method** because a slab is a circular washer of outer radius $R(x)$ and inner radius $r(x)$.

EXAMPLE 9 A Washer Cross-Section (Rotation About the x -Axis)

The region bounded by the curve $y = x^2 + 1$ and the line $y = -x + 3$ is revolved about the x -axis to generate a solid. Find the volume of the solid.

Solution

1. Draw the region and sketch a line segment across it perpendicular to the axis of revolution (the red segment in Figure 6.14).
2. Find the outer and inner radii of the washer that would be swept out by the line segment if it were revolved about the x -axis along with the region.

These radii are the distances of the ends of the line segment from the axis of revolution (Figure 6.14).

$$\text{Outer radius: } R(x) = -x + 3$$

$$\text{Inner radius: } r(x) = x^2 + 1$$

3. Find the limits of integration by finding the x -coordinates of the intersection points of the curve and line in Figure 6.14a.

$$x^2 + 1 = -x + 3$$

$$x^2 + x - 2 = 0$$

$$(x + 2)(x - 1) = 0$$

$$x = -2, \quad x = 1$$

4. Evaluate the volume integral.

$$\begin{aligned} V &= \int_a^b \pi([R(x)]^2 - [r(x)]^2) dx \\ &= \int_{-2}^1 \pi((-x + 3)^2 - (x^2 + 1)^2) dx && \text{Values from Steps 2 and 3} \\ &= \int_{-2}^1 \pi(8 - 6x - x^2 - x^4) dx \\ &= \pi \left[8x - 3x^2 - \frac{x^3}{3} - \frac{x^5}{5} \right]_{-2}^1 = \frac{117\pi}{5} \end{aligned}$$

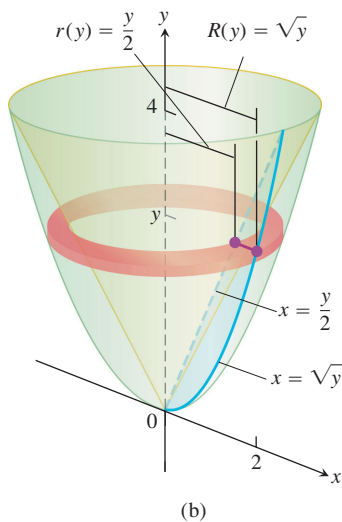
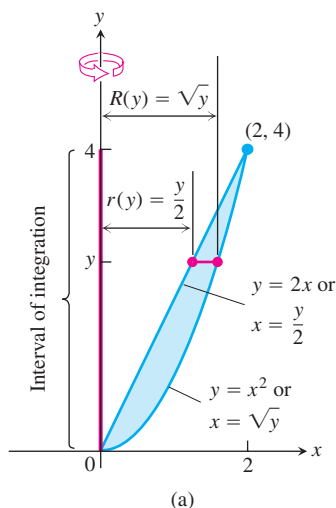


FIGURE 6.15 (a) The region being rotated about the y -axis, the washer radii, and limits of integration in Example 10. (b) The washer swept out by the line segment in part (a).

To find the volume of a solid formed by revolving a region about the y -axis, we use the same procedure as in Example 9, but integrate with respect to y instead of x . In this situation the line segment sweeping out a typical washer is perpendicular to the y -axis (the axis of revolution), and the outer and inner radii of the washer are functions of y .

EXAMPLE 10 A Washer Cross-Section (Rotation About the y -Axis)

The region bounded by the parabola $y = x^2$ and the line $y = 2x$ in the first quadrant is revolved about the y -axis to generate a solid. Find the volume of the solid.

Solution First we sketch the region and draw a line segment across it perpendicular to the axis of revolution (the y -axis). See Figure 6.15a.

The radii of the washer swept out by the line segment are $R(y) = \sqrt{y}$, $r(y) = y/2$ (Figure 6.15).

The line and parabola intersect at $y = 0$ and $y = 4$, so the limits of integration are $c = 0$ and $d = 4$. We integrate to find the volume:

$$\begin{aligned} V &= \int_c^d \pi([R(y)]^2 - [r(y)]^2) dy \\ &= \int_0^4 \pi \left(\left[\sqrt{y} \right]^2 - \left[\frac{y}{2} \right]^2 \right) dy \\ &= \pi \int_0^4 \left(y - \frac{y^2}{4} \right) dy = \pi \left[\frac{y^2}{2} - \frac{y^3}{12} \right]_0^4 = \frac{8}{3} \pi. \end{aligned}$$

Summary

In all of our volume examples, no matter how the cross-sectional area $A(x)$ of a typical slab is determined, the definition of volume as the definite integral $V = \int_a^b A(x) \, dx$ is the heart of the calculations we made.

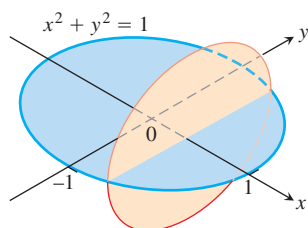
EXERCISES 6.1

Cross-Sectional Areas

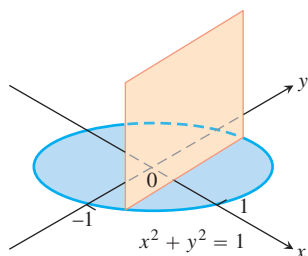
In Exercises 1 and 2, find a formula for the area $A(x)$ of the cross-sections of the solid perpendicular to the x -axis.

1. The solid lies between planes perpendicular to the x -axis at $x = -1$ and $x = 1$. In each case, the cross-sections perpendicular to the x -axis between these planes run from the semicircle $y = -\sqrt{1 - x^2}$ to the semicircle $y = \sqrt{1 - x^2}$.

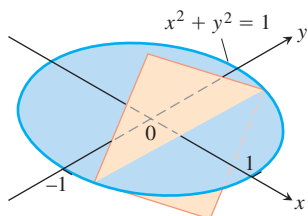
- a. The cross-sections are circular disks with diameters in the xy -plane.



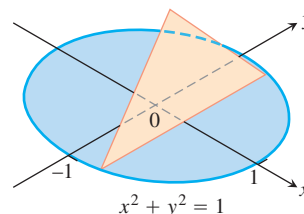
- b. The cross-sections are squares with bases in the xy -plane.



- c. The cross-sections are squares with diagonals in the xy -plane. (The length of a square's diagonal is $\sqrt{2}$ times the length of its sides.)

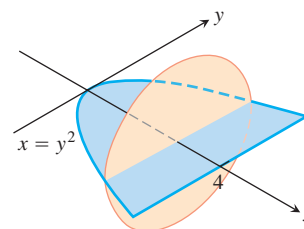


- d. The cross-sections are equilateral triangles with bases in the xy -plane.

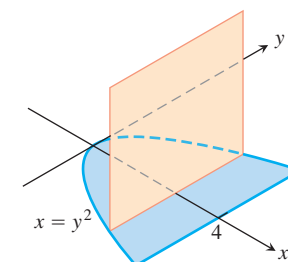


2. The solid lies between planes perpendicular to the x -axis at $x = 0$ and $x = 4$. The cross-sections perpendicular to the x -axis between these planes run from the parabola $y = -\sqrt{x}$ to the parabola $y = \sqrt{x}$.

- a. The cross-sections are circular disks with diameters in the xy -plane.



- b. The cross-sections are squares with bases in the xy -plane.

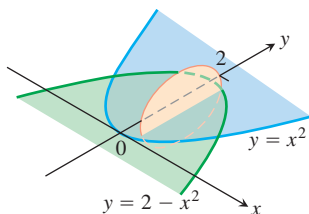


- c. The cross-sections are squares with diagonals in the xy -plane.
d. The cross-sections are equilateral triangles with bases in the xy -plane.

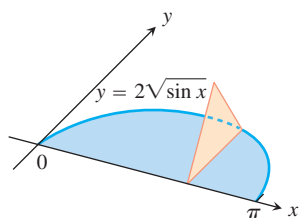
Volumes by Slicing

Find the volumes of the solids in Exercises 3–10.

3. The solid lies between planes perpendicular to the x -axis at $x = 0$ and $x = 4$. The cross-sections perpendicular to the axis on the interval $0 \leq x \leq 4$ are squares whose diagonals run from the parabola $y = -\sqrt{x}$ to the parabola $y = \sqrt{x}$.
4. The solid lies between planes perpendicular to the x -axis at $x = -1$ and $x = 1$. The cross-sections perpendicular to the x -axis are circular disks whose diameters run from the parabola $y = x^2$ to the parabola $y = 2 - x^2$.

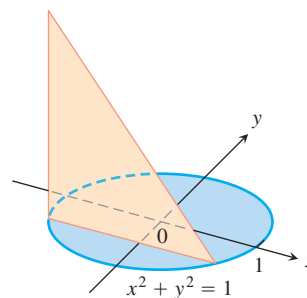


5. The solid lies between planes perpendicular to the x -axis at $x = -1$ and $x = 1$. The cross-sections perpendicular to the x -axis between these planes are squares whose bases run from the semicircle $y = -\sqrt{1 - x^2}$ to the semicircle $y = \sqrt{1 - x^2}$.
6. The solid lies between planes perpendicular to the x -axis at $x = -1$ and $x = 1$. The cross-sections perpendicular to the x -axis between these planes are squares whose diagonals run from the semicircle $y = -\sqrt{1 - x^2}$ to the semicircle $y = \sqrt{1 - x^2}$.
7. The base of a solid is the region between the curve $y = 2\sqrt{\sin x}$ and the interval $[0, \pi]$ on the x -axis. The cross-sections perpendicular to the x -axis are
 - a. equilateral triangles with bases running from the x -axis to the curve as shown in the figure.

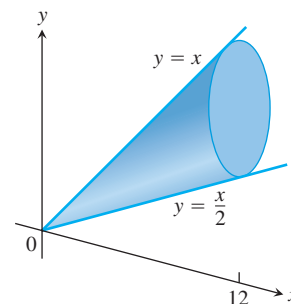


- b. squares with bases running from the x -axis to the curve.
8. The solid lies between planes perpendicular to the x -axis at $x = -\pi/3$ and $x = \pi/3$. The cross-sections perpendicular to the x -axis are
 - a. circular disks with diameters running from the curve $y = \tan x$ to the curve $y = \sec x$.
 - b. squares whose bases run from the curve $y = \tan x$ to the curve $y = \sec x$.
9. The solid lies between planes perpendicular to the y -axis at $y = 0$ and $y = 2$. The cross-sections perpendicular to the y -axis are circular disks with diameters running from the y -axis to the parabola $x = \sqrt{5}y^2$.

10. The base of the solid is the disk $x^2 + y^2 \leq 1$. The cross-sections by planes perpendicular to the y -axis between $y = -1$ and $y = 1$ are isosceles right triangles with one leg in the disk.



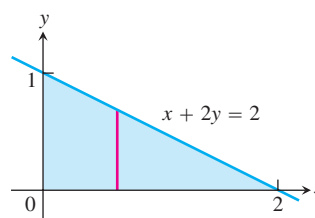
11. **A twisted solid** A square of side length s lies in a plane perpendicular to a line L . One vertex of the square lies on L . As this square moves a distance h along L , the square turns one revolution about L to generate a corkscrew-like column with square cross-sections.
 - a. Find the volume of the column.
 - b. What will the volume be if the square turns twice instead of once? Give reasons for your answer.
12. **Cavalieri's Principle** A solid lies between planes perpendicular to the x -axis at $x = 0$ and $x = 12$. The cross-sections by planes perpendicular to the x -axis are circular disks whose diameters run from the line $y = x/2$ to the line $y = x$ as shown in the accompanying figure. Explain why the solid has the same volume as a right circular cone with base radius 3 and height 12.



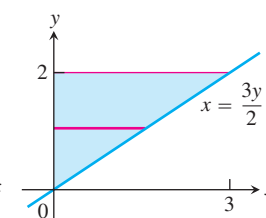
Volumes by the Disk Method

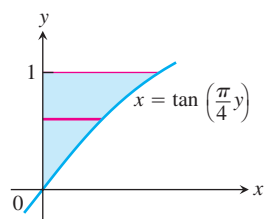
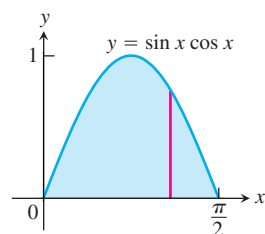
In Exercises 13–16, find the volume of the solid generated by revolving the shaded region about the given axis.

13. About the x -axis



14. About the y -axis



15. About the y -axis16. About the x -axis

Find the volumes of the solids generated by revolving the regions bounded by the lines and curves in Exercises 17–22 about the x -axis.

17. $y = x^2$, $y = 0$, $x = 2$ 18. $y = x^3$, $y = 0$, $x = 2$

19. $y = \sqrt{9 - x^2}$, $y = 0$ 20. $y = x - x^2$, $y = 0$

21. $y = \sqrt{\cos x}$, $0 \leq x \leq \pi/2$, $y = 0$, $x = 0$

22. $y = \sec x$, $y = 0$, $x = -\pi/4$, $x = \pi/4$

In Exercises 23 and 24, find the volume of the solid generated by revolving the region about the given line.

23. The region in the first quadrant bounded above by the line $y = \sqrt{2}$, below by the curve $y = \sec x \tan x$, and on the left by the y -axis, about the line $y = \sqrt{2}$

24. The region in the first quadrant bounded above by the line $y = 2$, below by the curve $y = 2 \sin x$, $0 \leq x \leq \pi/2$, and on the left by the y -axis, about the line $y = 2$

Find the volumes of the solids generated by revolving the regions bounded by the lines and curves in Exercises 25–30 about the y -axis.

25. The region enclosed by $x = \sqrt{5}y^2$, $x = 0$, $y = -1$, $y = 1$

26. The region enclosed by $x = y^{3/2}$, $x = 0$, $y = 2$

27. The region enclosed by $x = \sqrt{2 \sin 2y}$, $0 \leq y \leq \pi/2$, $x = 0$

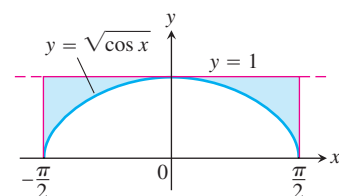
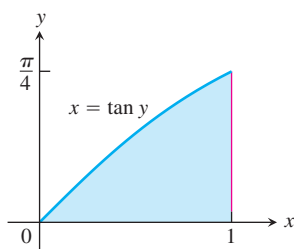
28. The region enclosed by $x = \sqrt{\cos(\pi y/4)}$, $-2 \leq y \leq 0$, $x = 0$

29. $x = 2/(y + 1)$, $x = 0$, $y = 0$, $y = 3$

30. $x = \sqrt{2y/(y^2 + 1)}$, $x = 0$, $y = 1$

Volumes by the Washer Method

Find the volumes of the solids generated by revolving the shaded regions in Exercises 31 and 32 about the indicated axes.

31. The x -axis32. The y -axis

Find the volumes of the solids generated by revolving the regions bounded by the lines and curves in Exercises 33–38 about the x -axis.

33. $y = x$, $y = 1$, $x = 0$ 34. $y = 2\sqrt{x}$, $y = 2$, $x = 0$

35. $y = x^2 + 1$, $y = x + 3$ 36. $y = 4 - x^2$, $y = 2 - x$

37. $y = \sec x$, $y = \sqrt{2}$, $-\pi/4 \leq x \leq \pi/4$

38. $y = \sec x$, $y = \tan x$, $x = 0$, $x = 1$

In Exercises 39–42, find the volume of the solid generated by revolving each region about the y -axis.

39. The region enclosed by the triangle with vertices $(1, 0)$, $(2, 1)$, and $(1, 1)$

40. The region enclosed by the triangle with vertices $(0, 1)$, $(1, 0)$, and $(1, 1)$

41. The region in the first quadrant bounded above by the parabola $y = x^2$, below by the x -axis, and on the right by the line $x = 2$

42. The region in the first quadrant bounded on the left by the circle $x^2 + y^2 = 3$, on the right by the line $x = \sqrt{3}$, and above by the line $y = \sqrt{3}$

In Exercises 43 and 44, find the volume of the solid generated by revolving each region about the given axis.

43. The region in the first quadrant bounded above by the curve $y = x^2$, below by the x -axis, and on the right by the line $x = 1$, about the line $x = -1$

44. The region in the second quadrant bounded above by the curve $y = -x^3$, below by the x -axis, and on the left by the line $x = -1$, about the line $x = -2$

Volumes of Solids of Revolution

45. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 2$ and $x = 0$ about

- a. the x -axis. b. the y -axis.
c. the line $y = 2$. d. the line $x = 4$.

46. Find the volume of the solid generated by revolving the triangular region bounded by the lines $y = 2x$, $y = 0$, and $x = 1$ about

- a. the line $x = 1$. b. the line $x = 2$.

47. Find the volume of the solid generated by revolving the region bounded by the parabola $y = x^2$ and the line $y = 1$ about

- a. the line $y = 1$. b. the line $y = 2$.
c. the line $y = -1$.

48. By integration, find the volume of the solid generated by revolving the triangular region with vertices $(0, 0)$, $(b, 0)$, $(0, h)$ about

- a. the x -axis. b. the y -axis.

Theory and Applications

49. **The volume of a torus** The disk $x^2 + y^2 \leq a^2$ is revolved about the line $x = b$ ($b > a$) to generate a solid shaped like a doughnut

and called a *torus*. Find its volume. (Hint: $\int_{-a}^a \sqrt{a^2 - y^2} dy = \pi a^2/2$, since it is the area of a semicircle of radius a .)

- 50. Volume of a bowl** A bowl has a shape that can be generated by revolving the graph of $y = x^2/2$ between $y = 0$ and $y = 5$ about the y -axis.

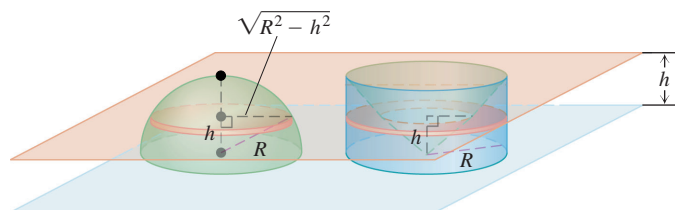
- Find the volume of the bowl.
- Related rates** If we fill the bowl with water at a constant rate of 3 cubic units per second, how fast will the water level in the bowl be rising when the water is 4 units deep?

51. Volume of a bowl

- A hemispherical bowl of radius a contains water to a depth h . Find the volume of water in the bowl.
- Related rates** Water runs into a sunken concrete hemispherical bowl of radius 5 m at the rate of $0.2 \text{ m}^3/\text{sec}$. How fast is the water level in the bowl rising when the water is 4 m deep?

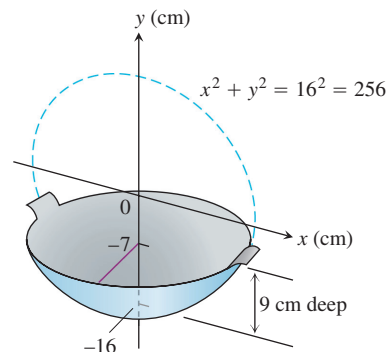
- 52.** Explain how you could estimate the volume of a solid of revolution by measuring the shadow cast on a table parallel to its axis of revolution by a light shining directly above it.

- 53. Volume of a hemisphere** Derive the formula $V = (2/3)\pi R^3$ for the volume of a hemisphere of radius R by comparing its cross-sections with the cross-sections of a solid right circular cylinder of radius R and height R from which a solid right circular cone of base radius R and height R has been removed as suggested by the accompanying figure.

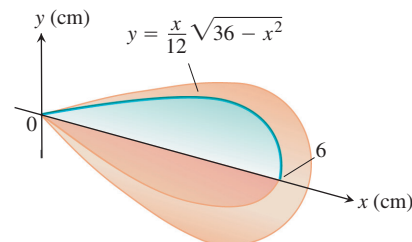


- 54. Volume of a cone** Use calculus to find the volume of a right circular cone of height h and base radius r .

- 55. Designing a wok** You are designing a wok frying pan that will be shaped like a spherical bowl with handles. A bit of experimentation at home persuades you that you can get one that holds about 3 L if you make it 9 cm deep and give the sphere a radius of 16 cm. To be sure, you picture the wok as a solid of revolution, as shown here, and calculate its volume with an integral. To the nearest cubic centimeter, what volume do you really get? (1 L = 1000 cm^3 .)



- 56. Designing a plumb bob** Having been asked to design a brass plumb bob that will weigh in the neighborhood of 190 g, you decide to shape it like the solid of revolution shown here. Find the plumb bob's volume. If you specify a brass that weighs 8.5 g/cm^3 , how much will the plumb bob weigh (to the nearest gram)?



- 57. Max-min** The arch $y = \sin x$, $0 \leq x \leq \pi$, is revolved about the line $y = c$, $0 \leq c \leq 1$, to generate the solid in Figure 6.16.

- Find the value of c that minimizes the volume of the solid. What is the minimum volume?
 - What value of c in $[0, 1]$ maximizes the volume of the solid?
- T** **c.** Graph the solid's volume as a function of c , first for $0 \leq c \leq 1$ and then on a larger domain. What happens to the volume of the solid as c moves away from $[0, 1]$? Does this make sense physically? Give reasons for your answers.

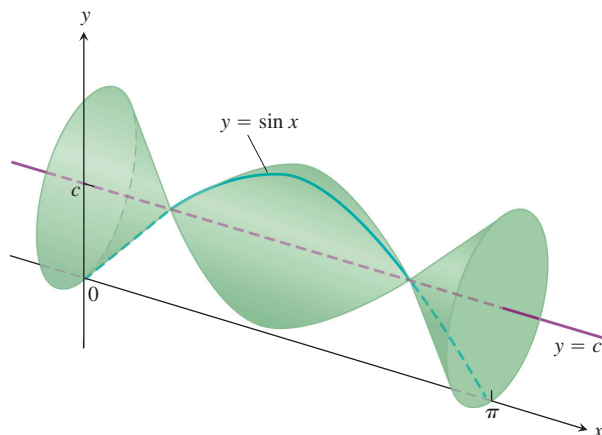


FIGURE 6.16

- 58. An auxiliary fuel tank** You are designing an auxiliary fuel tank that will fit under a helicopter's fuselage to extend its range. After some experimentation at your drawing board, you decide to shape the tank like the surface generated by revolving the curve $y = 1 - (x^2/16)$, $-4 \leq x \leq 4$, about the x -axis (dimensions in feet).
- How many cubic feet of fuel will the tank hold (to the nearest cubic foot)?
 - A cubic foot holds 7.481 gal. If the helicopter gets 2 mi to the gallon, how many additional miles will the helicopter be able to fly once the tank is installed (to the nearest mile)?

6.2 Volumes by Cylindrical Shells

In Section 6.1 we defined the volume of a solid S as the definite integral

$$V = \int_a^b A(x) \, dx,$$

where $A(x)$ is an integrable cross-sectional area of S from $x = a$ to $x = b$. The area $A(x)$ was obtained by slicing through the solid with a plane perpendicular to the x -axis. In this section we use the same integral definition for volume, but obtain the area by slicing through the solid in a different way. Now we slice through the solid using circular cylinders of increasing radii, like cookie cutters. We slice straight down through the solid perpendicular to the x -axis, with the axis of the cylinder parallel to the y -axis. The vertical axis of each cylinder is the same line, but the radii of the cylinders increase with each slice. In this way the solid S is sliced up into thin cylindrical shells of constant thickness that grow outward from their common axis, like circular tree rings. Unrolling a cylindrical shell shows that its volume is approximately that of a rectangular slab with area $A(x)$ and thickness Δx . This allows us to apply the same integral definition for volume as before. Before describing the method in general, let's look at an example to gain some insight.

EXAMPLE 1 Finding a Volume Using Shells

The region enclosed by the x -axis and the parabola $y = f(x) = 3x - x^2$ is revolved about the vertical line $x = -1$ to generate the shape of a solid (Figure 6.17). Find the volume of the solid.

Solution Using the washer method from Section 6.1 would be awkward here because we would need to express the x -values of the left and right branches of the parabola in terms

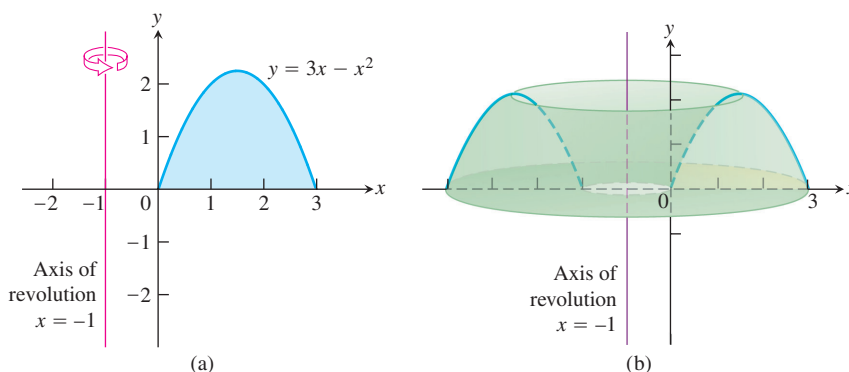


FIGURE 6.17 (a) The graph of the region in Example 1, before revolution. (b) The solid formed when the region in part (a) is revolved about the axis of revolution $x = -1$.

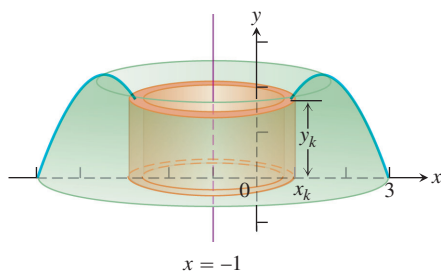


FIGURE 6.18 A cylindrical shell of height y_k obtained by rotating a vertical strip of thickness Δx about the line $x = -1$. The outer radius of the cylinder occurs at x_k , where the height of the parabola is $y_k = 3x_k - x_k^2$ (Example 1).

of y . (These x -values are the inner and outer radii for a typical washer, leading to complicated formulas.) Instead of rotating a horizontal strip of thickness Δy , we rotate a *vertical strip* of thickness Δx . This rotation produces a *cylindrical shell* of height y_k above a point x_k within the base of the vertical strip, and of thickness Δx . An example of a cylindrical shell is shown as the orange-shaded region in Figure 6.18. We can think of the cylindrical shell shown in the figure as approximating a slice of the solid obtained by cutting straight down through it, parallel to the axis of revolution, all the way around close to the inside hole. We then cut another cylindrical slice around the enlarged hole, then another, and so on, obtaining n cylinders. The radii of the cylinders gradually increase, and the heights of the cylinders follow the contour of the parabola: shorter to taller, then back to shorter (Figure 6.17a).

Each slice is sitting over a subinterval of the x -axis of length (width) Δx . Its radius is approximately $(1 + x_k)$, and its height is approximately $3x_k - x_k^2$. If we unroll the cylinder at x_k and flatten it out, it becomes (approximately) a rectangular slab with thickness Δx (Figure 6.19). The outer circumference of the k th cylinder is $2\pi \cdot \text{radius} = 2\pi(1 + x_k)$, and this is the length of the rolled-out rectangular slab. Its volume is approximated by that of a rectangular solid,

$$\begin{aligned}\Delta V_k &= \text{circumference} \times \text{height} \times \text{thickness} \\ &= 2\pi(1 + x_k) \cdot (3x_k - x_k^2) \cdot \Delta x.\end{aligned}$$

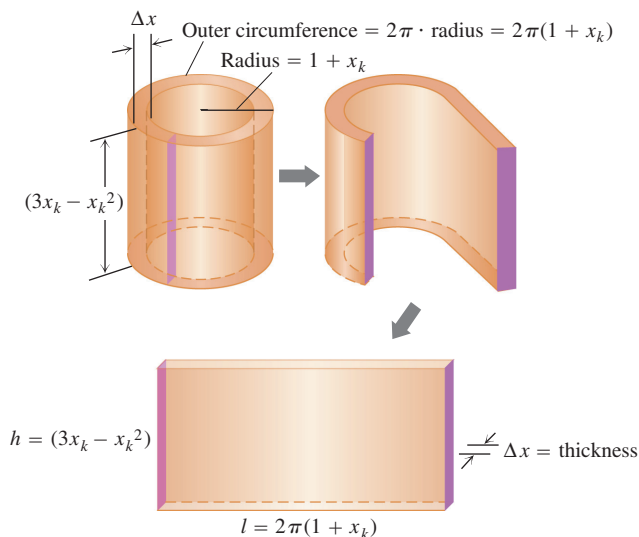


FIGURE 6.19 Imagine cutting and unrolling a cylindrical shell to get a flat (nearly) rectangular solid (Example 1).

Summing together the volumes ΔV_k of the individual cylindrical shells over the interval $[0, 3]$ gives the Riemann sum

$$\sum_{k=1}^n \Delta V_k = \sum_{k=1}^n 2\pi(x_k + 1)(3x_k - x_k^2) \Delta x.$$

Taking the limit as the thickness $\Delta x \rightarrow 0$ gives the volume integral

$$\begin{aligned}
 V &= \int_0^3 2\pi(x+1)(3x-x^2) dx \\
 &= \int_0^3 2\pi(3x^2+3x-x^3-x^2) dx \\
 &= 2\pi \int_0^3 (2x^2+3x-x^3) dx \\
 &= 2\pi \left[\frac{2}{3}x^3 + \frac{3}{2}x^2 - \frac{1}{4}x^4 \right]_0^3 \\
 &= \frac{45\pi}{2}.
 \end{aligned}$$

We now generalize the procedure used in Example 1.

The Shell Method

Suppose the region bounded by the graph of a nonnegative continuous function $y = f(x)$ and the x -axis over the finite closed interval $[a, b]$ lies to the right of the vertical line $x = L$ (Figure 6.20a). We assume $a \geq L$, so the vertical line may touch the region, but not pass through it. We generate a solid S by rotating this region about the vertical line L .

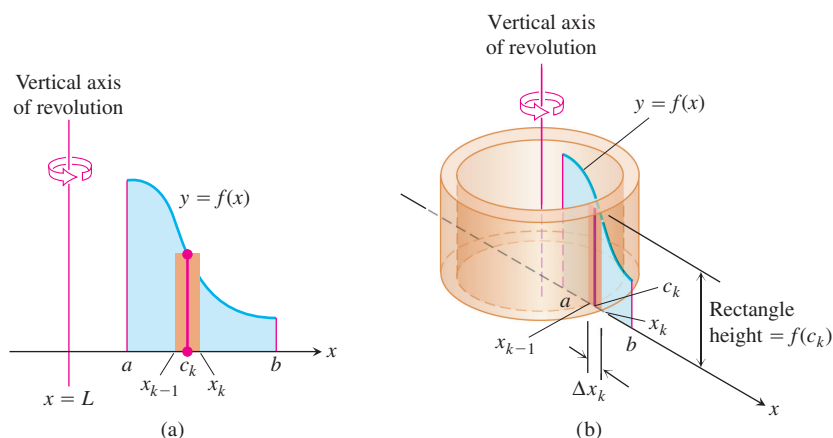


FIGURE 6.20 When the region shown in (a) is revolved about the vertical line $x = L$, a solid is produced which can be sliced into cylindrical shells. A typical shell is shown in (b).

Let P be a partition of the interval $[a, b]$ by the points $a = x_0 < x_1 < \cdots < x_n = b$, and let c_k be the midpoint of the k th subinterval $[x_{k-1}, x_k]$. We approximate the region in Figure 6.20a with rectangles based on this partition of $[a, b]$. A typical approximating rectangle has height $f(c_k)$ and width $\Delta x_k = x_k - x_{k-1}$. If this rectangle is rotated about the vertical line $x = L$, then a shell is swept out, as in Figure 6.20b. A formula from geometry tells us that the volume of the shell swept out by the rectangle is

$$\begin{aligned}
 \Delta V_k &= 2\pi \times \text{average shell radius} \times \text{shell height} \times \text{thickness} \\
 &= 2\pi \cdot (c_k - L) \cdot f(c_k) \cdot \Delta x_k.
 \end{aligned}$$

We approximate the volume of the solid S by summing the volumes of the shells swept out by the n rectangles based on P :

$$V \approx \sum_{k=1}^n \Delta V_k.$$

The limit of this Riemann sum as $\|P\| \rightarrow 0$ gives the volume of the solid as a definite integral:

$$\begin{aligned} V &= \int_a^b 2\pi(\text{shell radius})(\text{shell height}) \, dx. \\ &= \int_a^b 2\pi(x - L)f(x) \, dx. \end{aligned}$$

We refer to the variable of integration, here x , as the **thickness variable**. We use the first integral, rather than the second containing a formula for the integrand, to emphasize the *process* of the shell method. This will allow for rotations about a horizontal line L as well.

Shell Formula for Revolution About a Vertical Line

The volume of the solid generated by revolving the region between the x -axis and the graph of a continuous function $y = f(x) \geq 0$, $L \leq a \leq x \leq b$, about a vertical line $x = L$ is

$$V = \int_a^b 2\pi \left(\begin{matrix} \text{shell} \\ \text{radius} \end{matrix} \right) \left(\begin{matrix} \text{shell} \\ \text{height} \end{matrix} \right) dx.$$

EXAMPLE 2 Cylindrical Shells Revolving About the y -Axis

The region bounded by the curve $y = \sqrt{x}$, the x -axis, and the line $x = 4$ is revolved about the y -axis to generate a solid. Find the volume of the solid.

Solution Sketch the region and draw a line segment across it *parallel* to the axis of revolution (Figure 6.21a). Label the segment's height (shell height) and distance from the axis of revolution (shell radius). (We drew the shell in Figure 6.21b, but you need not do that.)

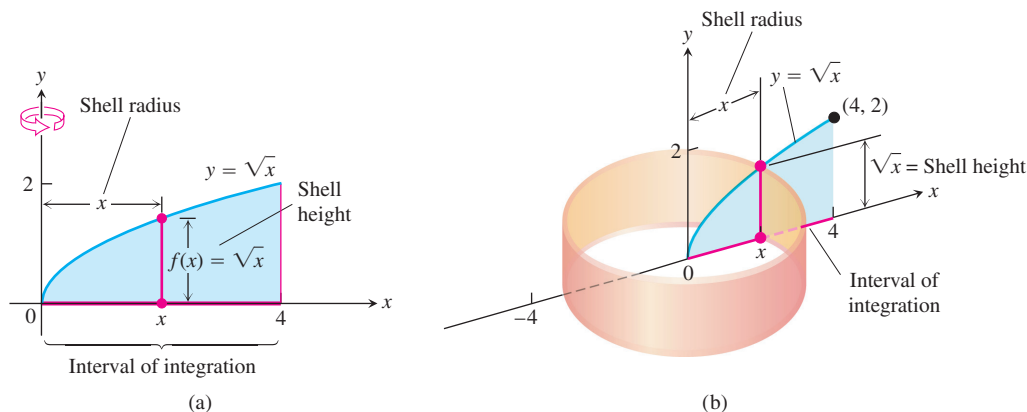


FIGURE 6.21 (a) The region, shell dimensions, and interval of integration in Example 2. (b) The shell swept out by the vertical segment in part (a) with a width Δx .

The shell thickness variable is x , so the limits of integration for the shell formula are $a = 0$ and $b = 4$ (Figure 6.20). The volume is then

$$\begin{aligned} V &= \int_a^b 2\pi \left(\begin{array}{c} \text{shell} \\ \text{radius} \end{array} \right) \left(\begin{array}{c} \text{shell} \\ \text{height} \end{array} \right) dx \\ &= \int_0^4 2\pi(x)(\sqrt{x}) dx \\ &= 2\pi \int_0^4 x^{3/2} dx = 2\pi \left[\frac{2}{5} x^{5/2} \right]_0^4 = \frac{128\pi}{5}. \end{aligned}$$

So far, we have used vertical axes of revolution. For horizontal axes, we replace the x 's with y 's.

EXAMPLE 3 Cylindrical Shells Revolving About the x -Axis

The region bounded by the curve $y = \sqrt{x}$, the x -axis, and the line $x = 4$ is revolved about the x -axis to generate a solid. Find the volume of the solid.

Solution Sketch the region and draw a line segment across it *parallel* to the axis of revolution (Figure 6.22a). Label the segment's length (shell height) and distance from the axis of revolution (shell radius). (We drew the shell in Figure 6.22b, but you need not do that.)

In this case, the shell thickness variable is y , so the limits of integration for the shell formula method are $a = 0$ and $b = 2$ (along the y -axis in Figure 6.22). The volume of the solid is

$$\begin{aligned} V &= \int_a^b 2\pi \left(\begin{array}{c} \text{shell} \\ \text{radius} \end{array} \right) \left(\begin{array}{c} \text{shell} \\ \text{height} \end{array} \right) dy \\ &= \int_0^2 2\pi(y)(4 - y^2) dy \\ &= \int_0^2 2\pi(4y - y^3) dy \\ &= 2\pi \left[2y^2 - \frac{y^4}{4} \right]_0^2 = 8\pi. \end{aligned}$$

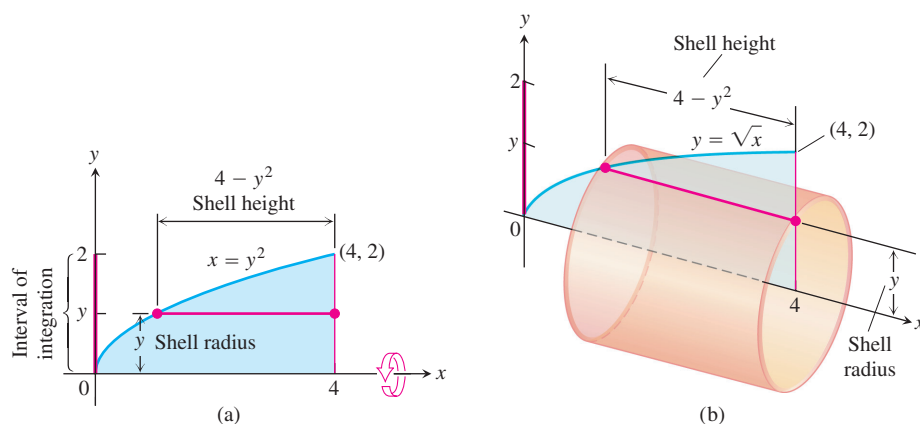


FIGURE 6.22 (a) The region, shell dimensions, and interval of integration in Example 3. (b) The shell swept out by the horizontal segment in part (a) with a width Δy .

Summary of the Shell Method

Regardless of the position of the axis of revolution (horizontal or vertical), the steps for implementing the shell method are these.

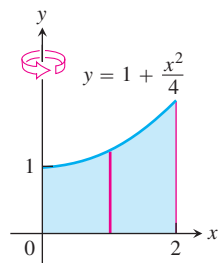
1. *Draw the region and sketch a line segment across it parallel to the axis of revolution. Label the segment's height or length (shell height) and distance from the axis of revolution (shell radius).*
2. *Find the limits of integration for the thickness variable.*
3. *Integrate the product 2π (shell radius) (shell height) with respect to the thickness variable (x or y) to find the volume.*

The shell method gives the same answer as the washer method when both are used to calculate the volume of a region. We do not prove that result here, but it is illustrated in Exercises 33 and 34. Both volume formulas are actually special cases of a general volume formula we look at in studying double and triple integrals in Chapter 15. That general formula also allows for computing volumes of solids other than those swept out by regions of revolution.

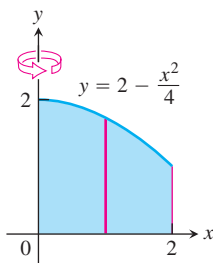
EXERCISES 6.2

In Exercises 1–6, use the shell method to find the volumes of the solids generated by revolving the shaded region about the indicated axis.

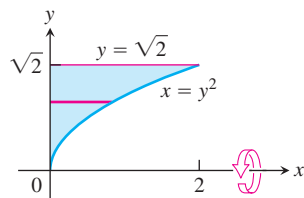
1.



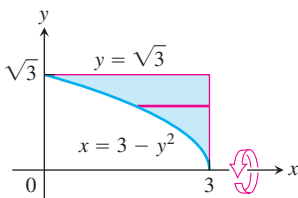
2.



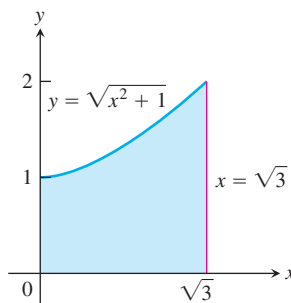
3.



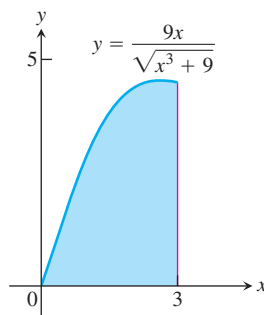
4.



5. The y-axis



6. The y-axis



Revolution About the y-Axis

Use the shell method to find the volumes of the solids generated by revolving the regions bounded by the curves and lines in Exercises 7–14 about the y-axis.

7. $y = x$, $y = -x/2$, $x = 2$

8. $y = 2x$, $y = x/2$, $x = 1$

9. $y = x^2$, $y = 2 - x$, $x = 0$, for $x \geq 0$

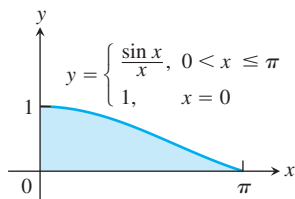
10. $y = 2 - x^2$, $y = x^2$, $x = 0$

11. $y = 2x - 1$, $y = \sqrt{x}$, $x = 0$

12. $y = 3/(2\sqrt{x})$, $y = 0$, $x = 1$, $x = 4$

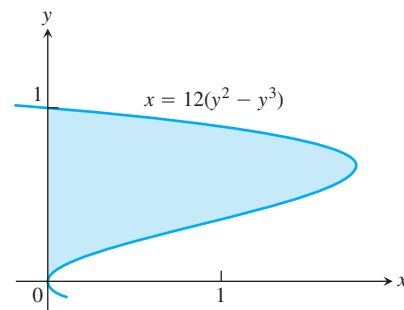
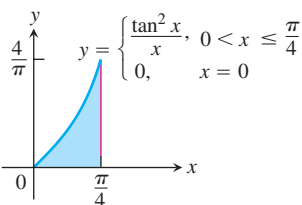
13. Let $f(x) = \begin{cases} (\sin x)/x, & 0 < x \leq \pi \\ 1, & x = 0 \end{cases}$

- Show that $xf(x) = \sin x$, $0 \leq x \leq \pi$.
- Find the volume of the solid generated by revolving the shaded region about the y -axis.

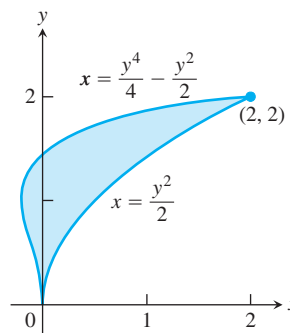


14. Let $g(x) = \begin{cases} (\tan x)^2/x, & 0 < x \leq \pi/4 \\ 0, & x = 0 \end{cases}$

- Show that $xg(x) = (\tan x)^2$, $0 \leq x \leq \pi/4$.
- Find the volume of the solid generated by revolving the shaded region about the y -axis.



- The x -axis
- The line $y = 2$
- The line $y = 5$
- The line $y = -5/8$



Revolution About the x -Axis

Use the shell method to find the volumes of the solids generated by revolving the regions bounded by the curves and lines in Exercises 15–22 about the x -axis.

- $x = \sqrt{y}$, $x = -y$, $y = 2$
- $x = y^2$, $x = -y$, $y = 2$, $y \geq 0$
- $x = 2y - y^2$, $x = 0$
- $x = 2y - y^2$, $x = y$
- $y = |x|$, $y = 1$
- $y = x$, $y = 2x$, $y = 2$
- $y = \sqrt{x}$, $y = 0$, $y = x - 2$
- $y = \sqrt{x}$, $y = 0$, $y = 2 - x$

Revolution About Horizontal Lines

In Exercises 23 and 24, use the shell method to find the volumes of the solids generated by revolving the shaded regions about the indicated axes.

- The x -axis
- The line $y = 1$
- The line $y = 8/5$
- The line $y = -2/5$

Comparing the Washer and Shell Models

For some regions, both the washer and shell methods work well for the solid generated by revolving the region about the coordinate axes, but this is not always the case. When a region is revolved about the y -axis, for example, and washers are used, we must integrate with respect to y . It may not be possible, however, to express the integrand in terms of y . In such a case, the shell method allows us to integrate with respect to x instead. Exercises 25 and 26 provide some insight.

- Compute the volume of the solid generated by revolving the region bounded by $y = x$ and $y = x^2$ about each coordinate axis using
 - the shell method.
 - the washer method.
- Compute the volume of the solid generated by revolving the triangular region bounded by the lines $2y = x + 4$, $y = x$, and $x = 0$ about
 - the x -axis using the washer method.
 - the y -axis using the shell method.
 - the line $x = 4$ using the shell method.
 - the line $y = 8$ using the washer method.

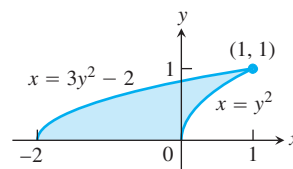
Choosing Shells or Washers

In Exercises 27–32, find the volumes of the solids generated by revolving the regions about the given axes. If you think it would be better to use washers in any given instance, feel free to do so.

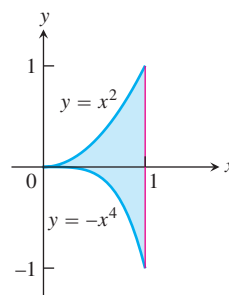
27. The triangle with vertices $(1, 1)$, $(1, 2)$, and $(2, 2)$ about
- the x -axis
 - the y -axis
 - the line $x = 10/3$
 - the line $y = 1$
28. The region bounded by $y = \sqrt{x}$, $y = 2$, $x = 0$ about
- the x -axis
 - the y -axis
 - the line $x = 4$
 - the line $y = 2$
29. The region in the first quadrant bounded by the curve $x = y - y^3$ and the y -axis about
- the x -axis
 - the line $y = 1$
30. The region in the first quadrant bounded by $x = y - y^3$, $x = 1$, and $y = 1$ about
- the x -axis
 - the y -axis
 - the line $x = 1$
 - the line $y = 1$
31. The region bounded by $y = \sqrt{x}$ and $y = x^2/8$ about
- the x -axis
 - the y -axis
32. The region bounded by $y = 2x - x^2$ and $y = x$ about
- the y -axis
 - the line $x = 1$
33. The region in the first quadrant that is bounded above by the curve $y = 1/x^{1/4}$, on the left by the line $x = 1/16$, and below by the line $y = 1$ is revolved about the x -axis to generate a solid. Find the volume of the solid by
- the washer method.
 - the shell method.
34. The region in the first quadrant that is bounded above by the curve $y = 1/\sqrt{x}$, on the left by the line $x = 1/4$, and below by the line $y = 1$ is revolved about the y -axis to generate a solid. Find the volume of the solid by
- the washer method.
 - the shell method.

Choosing Disks, Washers, or Shells

35. The region shown here is to be revolved about the x -axis to generate a solid. Which of the methods (disk, washer, shell) could you use to find the volume of the solid? How many integrals would be required in each case? Explain.



36. The region shown here is to be revolved about the y -axis to generate a solid. Which of the methods (disk, washer, shell) could you use to find the volume of the solid? How many integrals would be required in each case? Give reasons for your answers.



6.3

Lengths of Plane Curves

HISTORICAL BIOGRAPHY

Archimedes
(287–212 B.C.)

We know what is meant by the length of a straight line segment, but without calculus, we have no precise notion of the length of a general winding curve. The idea of approximating the length of a curve running from point A to point B by subdividing the curve into many pieces and joining successive points of division by straight line segments dates back to the ancient Greeks. Archimedes used this method to approximate the circumference of a circle by inscribing a polygon of n sides and then using geometry to compute its perimeter

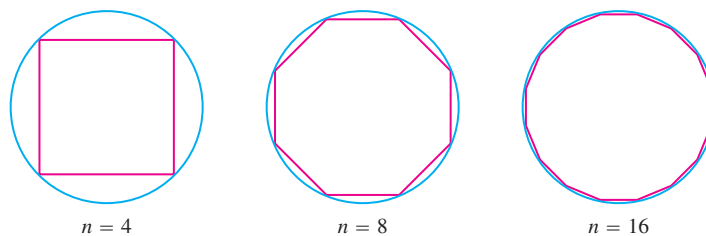


FIGURE 6.23 Archimedes used the perimeters of inscribed polygons to approximate the circumference of a circle. For $n = 96$ the approximation method gives $\pi \approx 3.14103$ as the circumference of the unit circle.

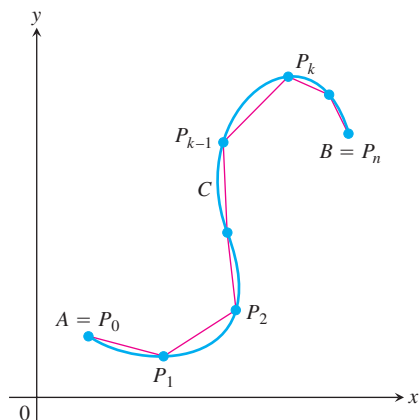


FIGURE 6.24 The curve C defined parametrically by the equations $x = f(t)$ and $y = g(t)$, $a \leq t \leq b$. The length of the curve from A to B is approximated by the sum of the lengths of the polygonal path (straight line segments) starting at $A = P_0$, then to P_1 , and so on, ending at $B = P_n$.

(Figure 6.23). The extension of this idea to a more general curve is displayed in Figure 6.24, and we now describe how that method works.

Length of a Parametrically Defined Curve

Let C be a curve given parametrically by the equations

$$x = f(t) \quad \text{and} \quad y = g(t), \quad a \leq t \leq b.$$

We assume the functions f and g have continuous derivatives on the interval $[a, b]$ that are not simultaneously zero. Such functions are said to be **continuously differentiable**, and the curve C defined by them is called a **smooth curve**. It may be helpful to imagine the curve as the path of a particle moving from point $A = (f(a), g(a))$ at time $t = a$ to point $B = (f(b), g(b))$ in Figure 6.24. We subdivide the path (or arc) AB into n pieces at points $A = P_0, P_1, P_2, \dots, P_n = B$. These points correspond to a partition of the interval $[a, b]$ by $a = t_0 < t_1 < t_2 < \dots < t_n = b$, where $P_k = (f(t_k), g(t_k))$. Join successive points of this subdivision by straight line segments (Figure 6.24). A representative line segment has length

$$\begin{aligned} L_k &= \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} \\ &= \sqrt{[f(t_k) - f(t_{k-1})]^2 + [g(t_k) - g(t_{k-1})]^2} \end{aligned}$$

(see Figure 6.25). If Δt_k is small, the length L_k is approximately the length of arc $P_{k-1}P_k$. By the Mean Value Theorem there are numbers t_k^* and t_k^{**} in $[t_{k-1}, t_k]$ such that

$$\begin{aligned} \Delta x_k &= f(t_k) - f(t_{k-1}) = f'(t_k^*) \Delta t_k, \\ \Delta y_k &= g(t_k) - g(t_{k-1}) = g'(t_k^{**}) \Delta t_k. \end{aligned}$$

Assuming the path from A to B is traversed exactly once as t increases from $t = a$ to $t = b$, with no doubling back or retracing, an intuitive approximation to the “length” of the curve AB is the sum of all the lengths L_k :

$$\begin{aligned} \sum_{k=1}^n L_k &= \sum_{k=1}^n \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} \\ &= \sum_{k=1}^n \sqrt{[f'(t_k^*)]^2 + [g'(t_k^{**})]^2} \Delta t_k. \end{aligned}$$

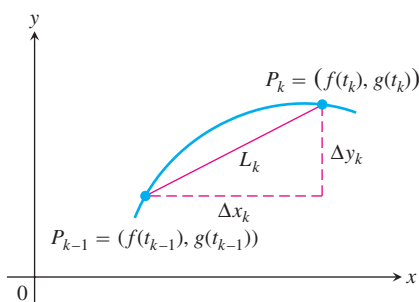


FIGURE 6.25 The arc $P_{k-1}P_k$ is approximated by the straight line segment shown here, which has length $L_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$.

Although this last sum on the right is not exactly a Riemann sum (because f' and g' are evaluated at different points), a theorem in advanced calculus guarantees its limit, as the norm of the partition tends to zero, to be the definite integral

$$\int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt.$$

Therefore, it is reasonable to define the length of the curve from A to B as this integral.

DEFINITION Length of a Parametric Curve

If a curve C is defined parametrically by $x = f(t)$ and $y = g(t)$, $a \leq t \leq b$, where f' and g' are continuous and not simultaneously zero on $[a, b]$, and C is traversed exactly once as t increases from $t = a$ to $t = b$, then **the length of C** is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt.$$

A smooth curve C does not double back or reverse the direction of motion over the time interval $[a, b]$ since $(f')^2 + (g')^2 > 0$ throughout the interval.

If $x = f(t)$ and $y = g(t)$, then using the Leibniz notation we have the following result for arc length:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt. \quad (1)$$

What if there are two different parametrizations for a curve C whose length we want to find; does it matter which one we use? The answer, from advanced calculus, is no, as long as the parametrization we choose meets the conditions stated in the definition of the length of C (see Exercise 29).

EXAMPLE 1 The Circumference of a Circle

Find the length of the circle of radius r defined parametrically by

$$x = r \cos t \quad \text{and} \quad y = r \sin t, \quad 0 \leq t \leq 2\pi.$$

Solution As t varies from 0 to 2π , the circle is traversed exactly once, so the circumference is

$$L = \int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

We find

$$\frac{dx}{dt} = -r \sin t, \quad \frac{dy}{dt} = r \cos t$$

and

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = r^2(\sin^2 t + \cos^2 t) = r^2.$$

So

$$L = \int_0^{2\pi} \sqrt{r^2} dt = r [t]_0^{2\pi} = 2\pi r.$$

EXAMPLE 2 Applying the Parametric Formula for Length of a Curve

Find the length of the astroid (Figure 6.26)

$$x = \cos^3 t, \quad y = \sin^3 t, \quad 0 \leq t \leq 2\pi.$$

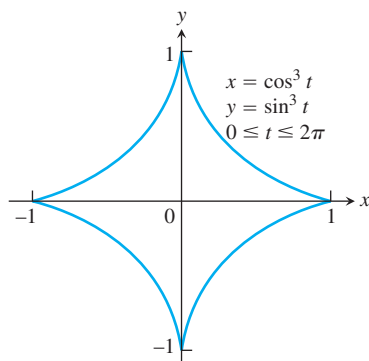


FIGURE 6.26 The astroid in Example 2.

Solution Because of the curve's symmetry with respect to the coordinate axes, its length is four times the length of the first-quadrant portion. We have

$$\begin{aligned} x &= \cos^3 t, & y &= \sin^3 t \\ \left(\frac{dx}{dt}\right)^2 &= [3\cos^2 t(-\sin t)]^2 = 9\cos^4 t \sin^2 t \\ \left(\frac{dy}{dt}\right)^2 &= [3\sin^2 t(\cos t)]^2 = 9\sin^4 t \cos^2 t \\ \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} &= \sqrt{9\cos^2 t \sin^2 t (\underbrace{\cos^2 t + \sin^2 t}_1)} \\ &= \sqrt{9\cos^2 t \sin^2 t} \\ &= 3|\cos t \sin t| \\ &= 3\cos t \sin t. \end{aligned}$$

$\cos t \sin t \geq 0$ for
 $0 \leq t \leq \pi/2$

Therefore,

$$\begin{aligned} \text{Length of first-quadrant portion} &= \int_0^{\pi/2} 3\cos t \sin t dt \\ &= \frac{3}{2} \int_0^{\pi/2} \sin 2t dt \\ &= -\frac{3}{4} \cos 2t \Big|_0^{\pi/2} = \frac{3}{2}. \end{aligned}$$

The length of the astroid is four times this: $4(3/2) = 6$.

HISTORICAL BIOGRAPHY

Gregory St. Vincent
(1584–1667)

Length of a Curve $y = f(x)$

Given a continuously differentiable function $y = f(x)$, $a \leq x \leq b$, we can assign $x = t$ as a parameter. The graph of the function f is then the curve C defined parametrically by

$$x = t \quad \text{and} \quad y = f(t), \quad a \leq t \leq b,$$

a special case of what we considered before. Then,

$$\frac{dx}{dt} = 1 \quad \text{and} \quad \frac{dy}{dt} = f'(t).$$

From our calculations in Section 3.5, we have

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = f'(t)$$

giving

$$\begin{aligned} \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= 1 + [f'(t)]^2 \\ &= 1 + \left(\frac{dy}{dx}\right)^2 \\ &= 1 + [f'(x)]^2. \end{aligned}$$

Substitution into Equation (1) gives the arc length formula for the graph of $y = f(x)$.

Formula for the Length of $y = f(x)$, $a \leq x \leq b$

If f is continuously differentiable on the closed interval $[a, b]$, the length of the curve (graph) $y = f(x)$ from $x = a$ to $x = b$ is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b \sqrt{1 + [f'(x)]^2} dx. \quad (2)$$

EXAMPLE 3 Applying the Arc Length Formula for a Graph

Find the length of the curve

$$y = \frac{4\sqrt{2}}{3}x^{3/2} - 1, \quad 0 \leq x \leq 1.$$

Solution We use Equation (2) with $a = 0$, $b = 1$, and

$$\begin{aligned} y &= \frac{4\sqrt{2}}{3}x^{3/2} - 1 \\ \frac{dy}{dx} &= \frac{4\sqrt{2}}{3} \cdot \frac{3}{2}x^{1/2} = 2\sqrt{2}x^{1/2} \\ \left(\frac{dy}{dx}\right)^2 &= (2\sqrt{2}x^{1/2})^2 = 8x. \end{aligned}$$

The length of the curve from $x = 0$ to $x = 1$ is

$$\begin{aligned} L &= \int_0^1 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_0^1 \sqrt{1 + 8x} dx \\ &= \frac{2}{3} \cdot \frac{1}{8} (1 + 8x)^{3/2} \Big|_0^1 = \frac{13}{6}. \end{aligned}$$

Eq. (2) with
 $a = 0$, $b = 1$
Let $u = 1 + 8x$,
integrate, and
replace u by
 $1 + 8x$. ■

Dealing with Discontinuities in dy/dx

At a point on a curve where dy/dx fails to exist, dx/dy may exist and we may be able to find the curve's length by expressing x as a function of y and applying the following analogue of Equation (2):

Formula for the Length of $x = g(y)$, $c \leq y \leq d$

If g is continuously differentiable on $[c, d]$, the length of the curve $x = g(y)$ from $y = c$ to $y = d$ is

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_c^d \sqrt{1 + [g'(y)]^2} dy. \quad (3)$$

EXAMPLE 4 Length of a Graph Which Has a Discontinuity in dy/dx

Find the length of the curve $y = (x/2)^{2/3}$ from $x = 0$ to $x = 2$.

Solution The derivative

$$\frac{dy}{dx} = \frac{2}{3} \left(\frac{x}{2}\right)^{-1/3} \left(\frac{1}{2}\right) = \frac{1}{3} \left(\frac{2}{x}\right)^{1/3}$$

is not defined at $x = 0$, so we cannot find the curve's length with Equation (2).

We therefore rewrite the equation to express x in terms of y :

$$\begin{aligned} y &= \left(\frac{x}{2}\right)^{2/3} \\ y^{3/2} &= \frac{x}{2} && \text{Raise both sides to the power } 3/2. \\ x &= 2y^{3/2}. && \text{Solve for } x. \end{aligned}$$

From this we see that the curve whose length we want is also the graph of $x = 2y^{3/2}$ from $y = 0$ to $y = 1$ (Figure 6.27).

The derivative

$$\frac{dx}{dy} = 2 \left(\frac{3}{2}\right) y^{1/2} = 3y^{1/2}$$

is continuous on $[0, 1]$. We may therefore use Equation (3) to find the curve's length:

$$\begin{aligned} L &= \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_0^1 \sqrt{1 + 9y} dy && \text{Eq. (3) with } c = 0, d = 1. \\ &= \frac{1}{9} \cdot \frac{2}{3} (1 + 9y)^{3/2} \Big|_0^1 && \text{Let } u = 1 + 9y, \\ &= \frac{2}{27} (10\sqrt{10} - 1) \approx 2.27. && du/9 = dy, \text{ integrate, and substitute back.} \end{aligned}$$

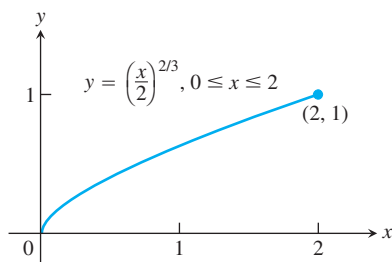


FIGURE 6.27 The graph of $y = (x/2)^{2/3}$ from $x = 0$ to $x = 2$ is also the graph of $x = 2y^{3/2}$ from $y = 0$ to $y = 1$ (Example 4).

HISTORICAL BIOGRAPHY

James Gregory
(1638–1675)

The Short Differential Formula

Equation (1) is frequently written in terms of differentials in place of derivatives. This is done formally by writing $(dt)^2$ under the radical in place of the dt outside the radical, and then writing

$$\left(\frac{dx}{dt}\right)^2 (dt)^2 = \left(\frac{dx}{dt} dt\right)^2 = (dx)^2$$

and

$$\left(\frac{dy}{dt}\right)^2 (dt)^2 = \left(\frac{dy}{dt} dt\right)^2 = (dy)^2.$$

It is also customary to eliminate the parentheses in $(dx)^2$ and write dx^2 instead, so that Equation (1) is written

$$L = \int \sqrt{dx^2 + dy^2}. \quad (4)$$

We can think of these differentials as a way to summarize and simplify the properties of integrals. Differentials are given a precise mathematical definition in a more advanced text.

To do an integral computation, dx and dy must both be expressed in terms of one and the same variable, and appropriate limits must be supplied in Equation (4).

A useful way to remember Equation (4) is to write

$$ds = \sqrt{dx^2 + dy^2} \quad (5)$$

and treat ds as the differential of arc length, which can be integrated between appropriate limits to give the total length of a curve. Figure 6.28a gives the exact interpretation of ds corresponding to Equation (5). Figure 6.28b is not strictly accurate but is to be thought of as a simplified approximation of Figure 6.28a.

With Equation (5) in mind, the quickest way to recall the formulas for arc length is to remember the equation

$$\text{Arc length} = \int ds.$$

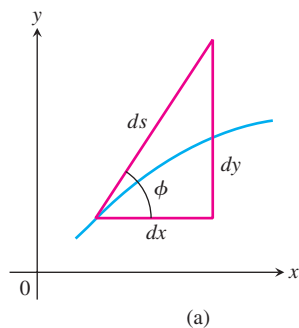
If we write $L = \int ds$ and have the graph of $y = f(x)$, we can rewrite Equation (5) to get

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{dx^2 + \frac{dy^2}{dx^2} dx^2} = \sqrt{1 + \frac{dy^2}{dx^2}} dx = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx,$$

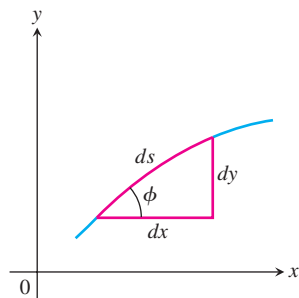
resulting in Equation (2). If we have instead $x = g(y)$, we rewrite Equation (5)

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{dy^2 + \frac{dx^2}{dy^2} dy^2} = \sqrt{1 + \frac{dx^2}{dy^2}} dy = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy,$$

and obtain Equation (3).



(a)



(b)

FIGURE 6.28 Diagrams for remembering the equation $ds = \sqrt{dx^2 + dy^2}$.

EXERCISES 6.3

Lengths of Parametrized Curves

Find the lengths of the curves in Exercises 1–6.

1. $x = 1 - t$, $y = 2 + 3t$, $-2/3 \leq t \leq 1$
2. $x = \cos t$, $y = t + \sin t$, $0 \leq t \leq \pi$
3. $x = t^3$, $y = 3t^2/2$, $0 \leq t \leq \sqrt{3}$
4. $x = t^2/2$, $y = (2t + 1)^{3/2}/3$, $0 \leq t \leq 4$
5. $x = (2t + 3)^{3/2}/3$, $y = t + t^2/2$, $0 \leq t \leq 3$
6. $x = 8 \cos t + 8t \sin t$, $y = 8 \sin t - 8t \cos t$, $0 \leq t \leq \pi/2$

Finding Lengths of Curves

Find the lengths of the curves in Exercises 7–16. If you have a grapher, you may want to graph these curves to see what they look like.

7. $y = (1/3)(x^2 + 2)^{3/2}$ from $x = 0$ to $x = 3$
8. $y = x^{3/2}$ from $x = 0$ to $x = 4$
9. $x = (y^3/3) + 1/(4y)$ from $y = 1$ to $y = 3$
(Hint: $1 + (dx/dy)^2$ is a perfect square.)
10. $x = (y^{3/2}/3) - y^{1/2}$ from $y = 1$ to $y = 9$
(Hint: $1 + (dx/dy)^2$ is a perfect square.)
11. $x = (y^4/4) + 1/(8y^2)$ from $y = 1$ to $y = 2$
(Hint: $1 + (dx/dy)^2$ is a perfect square.)
12. $x = (y^3/6) + 1/(2y)$ from $y = 2$ to $y = 3$
(Hint: $1 + (dx/dy)^2$ is a perfect square.)
13. $y = (3/4)x^{4/3} - (3/8)x^{2/3} + 5$, $1 \leq x \leq 8$
14. $y = (x^3/3) + x^2 + x + 1/(4x + 4)$, $0 \leq x \leq 2$
15. $x = \int_0^y \sqrt{\sec^4 t - 1} dt$, $-\pi/4 \leq y \leq \pi/4$
16. $y = \int_{-2}^x \sqrt{3t^4 - 1} dt$, $-2 \leq x \leq -1$

T Finding Integrals for Lengths of Curves

In Exercises 17–24, do the following.

- a. Set up an integral for the length of the curve.
 - b. Graph the curve to see what it looks like.
 - c. Use your grapher's or computer's integral evaluator to find the curve's length numerically.
17. $y = x^2$, $-1 \leq x \leq 2$
 18. $y = \tan x$, $-\pi/3 \leq x \leq 0$
 19. $x = \sin y$, $0 \leq y \leq \pi$
 20. $x = \sqrt{1 - y^2}$, $-1/2 \leq y \leq 1/2$
 21. $y^2 + 2y = 2x + 1$ from $(-1, -1)$ to $(7, 3)$
 22. $y = \sin x - x \cos x$, $0 \leq x \leq \pi$

$$23. y = \int_0^x \tan t dt, \quad 0 \leq x \leq \pi/6$$

$$24. x = \int_0^y \sqrt{\sec^2 t - 1} dt, \quad -\pi/3 \leq y \leq \pi/4$$

Theory and Applications

25. Is there a smooth (continuously differentiable) curve $y = f(x)$ whose length over the interval $0 \leq x \leq a$ is always $\sqrt{2}a$? Give reasons for your answer.

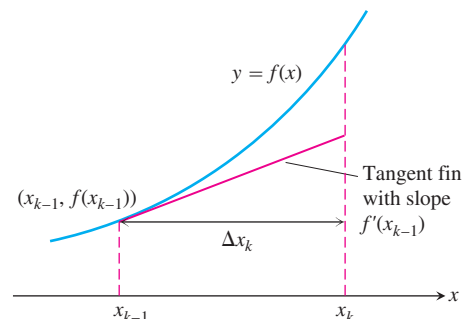
26. **Using tangent fins to derive the length formula for curves** Assume that f is smooth on $[a, b]$ and partition the interval $[a, b]$ in the usual way. In each subinterval $[x_{k-1}, x_k]$, construct the *tangent fin* at the point $(x_{k-1}, f(x_{k-1}))$, as shown in the accompanying figure.

a. Show that the length of the k th tangent fin over the interval $[x_{k-1}, x_k]$ equals $\sqrt{(\Delta x_k)^2 + (f'(x_{k-1}) \Delta x_k)^2}$.

b. Show that

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n (\text{length of } k\text{th tangent fin}) = \int_a^b \sqrt{1 + (f'(x))^2} dx,$$

which is the length L of the curve $y = f(x)$ from a to b .



27. a. Find a curve through the point $(1, 1)$ whose length integral is

$$L = \int_1^4 \sqrt{1 + \frac{1}{4x}} dx.$$

b. How many such curves are there? Give reasons for your answer.

28. a. Find a curve through the point $(0, 1)$ whose length integral is

$$L = \int_1^2 \sqrt{1 + \frac{1}{y^4}} dy.$$

b. How many such curves are there? Give reasons for your answer.

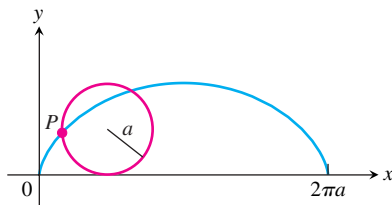
29. **Length is independent of parametrization** To illustrate the fact that the numbers we get for length do not depend on the way

we parametrize our curves (except for the mild restrictions preventing doubling back mentioned earlier), calculate the length of the semicircle $y = \sqrt{1 - x^2}$ with these two different parametrizations:

a. $x = \cos 2t, \quad y = \sin 2t, \quad 0 \leq t \leq \pi/2$

b. $x = \sin \pi t, \quad y = \cos \pi t, \quad -1/2 \leq t \leq 1/2$

30. Find the length of one arch of the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$, $0 \leq \theta \leq 2\pi$, shown in the accompanying figure. A **cycloid** is the curve traced out by a point P on the circumference of a circle rolling along a straight line, such as the x -axis.



COMPUTER EXPLORATIONS

In Exercises 31–36, use a CAS to perform the following steps for the given curve over the closed interval.

- Plot the curve together with the polygonal path approximations for $n = 2, 4, 8$ partition points over the interval. (See Figure 6.24.)
- Find the corresponding approximation to the length of the curve by summing the lengths of the line segments.
- Evaluate the length of the curve using an integral. Compare your approximations for $n = 2, 4, 8$ with the actual length given by the integral. How does the actual length compare with the approximations as n increases? Explain your answer.

31. $f(x) = \sqrt{1 - x^2}, \quad -1 \leq x \leq 1$

32. $f(x) = x^{1/3} + x^{2/3}, \quad 0 \leq x \leq 2$

33. $f(x) = \sin(\pi x^2), \quad 0 \leq x \leq \sqrt{2}$

34. $f(x) = x^2 \cos x, \quad 0 \leq x \leq \pi$

35. $f(x) = \frac{x-1}{4x^2+1}, \quad -\frac{1}{2} \leq x \leq 1$

36. $f(x) = x^3 - x^2, \quad -1 \leq x \leq 1$

37. $x = \frac{1}{3}t^3, \quad y = \frac{1}{2}t^2, \quad 0 \leq t \leq 1$

38. $x = 2t^3 - 16t^2 + 25t + 5, \quad y = t^2 + t - 3, \quad 0 \leq t \leq 6$

39. $x = t - \cos t, \quad y = 1 + \sin t, \quad -\pi \leq t \leq \pi$

40. $x = e^t \cos t, \quad y = e^t \sin t, \quad 0 \leq t \leq \pi$

6.4

Moments and Centers of Mass

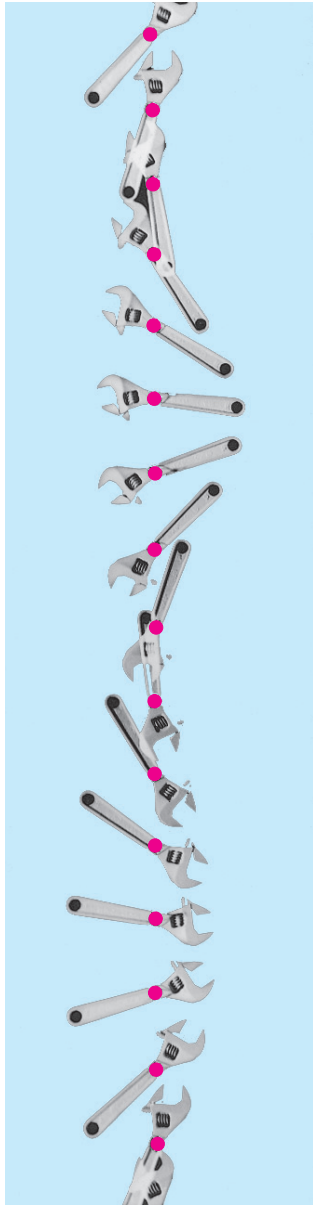
Many structures and mechanical systems behave as if their masses were concentrated at a single point, called the *center of mass* (Figure 6.29). It is important to know how to locate this point, and doing so is basically a mathematical enterprise. For the moment, we deal with one- and two-dimensional objects. Three-dimensional objects are best done with the multiple integrals of Chapter 15.

Masses Along a Line

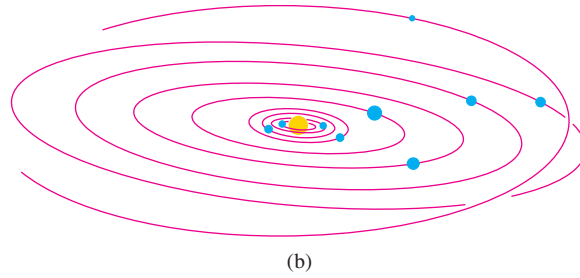
We develop our mathematical model in stages. The first stage is to imagine masses m_1 , m_2 , and m_3 on a rigid x -axis supported by a fulcrum at the origin.



The resulting system might balance, or it might not. It depends on how large the masses are and how they are arranged.



(a)



(b)

FIGURE 6.29 (a) The motion of this wrench gliding on ice seems haphazard until we notice that the wrench is simply turning about its center of mass as the center glides in a straight line. (b) The planets, asteroids, and comets of our solar system revolve about their collective center of mass. (It lies inside the sun.)

Each mass m_k exerts a downward force $m_k g$ (the weight of m_k) equal to the magnitude of the mass times the acceleration of gravity. Each of these forces has a tendency to turn the axis about the origin, the way you turn a seesaw. This turning effect, called a **torque**, is measured by multiplying the force $m_k g$ by the signed distance x_k from the point of application to the origin. Masses to the left of the origin exert negative (counterclockwise) torque. Masses to the right of the origin exert positive (clockwise) torque.

The sum of the torques measures the tendency of a system to rotate about the origin. This sum is called the **system torque**.

$$\text{System torque} = m_1 g x_1 + m_2 g x_2 + m_3 g x_3 \quad (1)$$

The system will balance if and only if its torque is zero.

If we factor out the g in Equation (1), we see that the system torque is

$$\underbrace{g}_{\text{a feature of the environment}} \cdot \underbrace{(m_1 x_1 + m_2 x_2 + m_3 x_3)}_{\text{a feature of the system}}$$

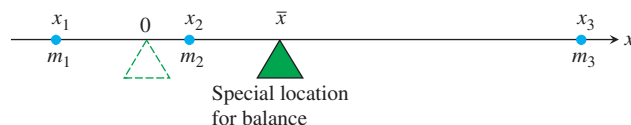
Thus, the torque is the product of the gravitational acceleration g , which is a feature of the environment in which the system happens to reside, and the number $(m_1 x_1 + m_2 x_2 + m_3 x_3)$, which is a feature of the system itself, a constant that stays the same no matter where the system is placed.

The number $(m_1 x_1 + m_2 x_2 + m_3 x_3)$ is called the **moment of the system about the origin**. It is the sum of the **moments** $m_1 x_1$, $m_2 x_2$, $m_3 x_3$ of the individual masses.

$$M_0 = \text{Moment of system about origin} = \sum m_k x_k.$$

(We shift to sigma notation here to allow for sums with more terms.)

We usually want to know where to place the fulcrum to make the system balance, that is, at what point $\bar{x}$ to place it to make the torques add to zero.



The torque of each mass about the fulcrum in this special location is

$$\begin{aligned}\text{Torque of } m_k \text{ about } \bar{x} &= \left(\begin{array}{c} \text{signed distance} \\ \text{of } m_k \text{ from } \bar{x} \end{array} \right) \left(\begin{array}{c} \text{downward} \\ \text{force} \end{array} \right) \\ &= (x_k - \bar{x})m_k g.\end{aligned}$$

When we write the equation that says that the sum of these torques is zero, we get an equation we can solve for $\bar{x}$:

$$\begin{aligned}\sum (x_k - \bar{x})m_k g &= 0 && \text{Sum of the torques equals zero} \\ g \sum (x_k - \bar{x})m_k &= 0 && \text{Constant Multiple Rule for Sums} \\ \sum (m_k x_k - \bar{x}m_k) &= 0 && g \text{ divided out, } m_k \text{ distributed} \\ \sum m_k x_k - \sum \bar{x}m_k &= 0 && \text{Difference Rule for Sums} \\ \sum m_k x_k &= \bar{x} \sum m_k && \text{Rearranged, Constant Multiple Rule again} \\ \bar{x} &= \frac{\sum m_k x_k}{\sum m_k}. && \text{Solved for } \bar{x}\end{aligned}$$

This last equation tells us to find $\bar{x}$ by dividing the system's moment about the origin by the system's total mass:

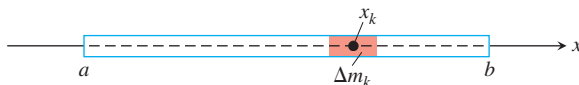
$$\bar{x} = \frac{\sum m_k x_k}{\sum m_k} = \frac{\text{system moment about origin}}{\text{system mass}}.$$

The point $\bar{x}$ is called the system's **center of mass**.

Wires and Thin Rods

In many applications, we want to know the center of mass of a rod or a thin strip of metal. In cases like these where we can model the distribution of mass with a continuous function, the summation signs in our formulas become integrals in a manner we now describe.

Imagine a long, thin strip lying along the x -axis from $x = a$ to $x = b$ and cut into small pieces of mass Δm_k by a partition of the interval $[a, b]$. Choose x_k to be any point in the k th subinterval of the partition.



The k th piece is Δx_k units long and lies approximately x_k units from the origin. Now observe three things.

First, the strip's center of mass $\bar{x}$ is nearly the same as that of the system of point masses we would get by attaching each mass Δm_k to the point x_k :

$$\bar{x} \approx \frac{\text{system moment}}{\text{system mass}}.$$

Density

A material's density is its mass per unit volume. In practice, however, we tend to use units we can conveniently measure. For wires, rods, and narrow strips, we use mass per unit length. For flat sheets and plates, we use mass per unit area.

Second, the moment of each piece of the strip about the origin is approximately $x_k \Delta m_k$, so the system moment is approximately the sum of the $x_k \Delta m_k$:

$$\text{System moment} \approx \sum x_k \Delta m_k.$$

Third, if the density of the strip at x_k is $\delta(x_k)$, expressed in terms of mass per unit length and if δ is continuous, then Δm_k is approximately equal to $\delta(x_k) \Delta x_k$ (mass per unit length times length):

$$\Delta m_k \approx \delta(x_k) \Delta x_k.$$

Combining these three observations gives

$$\bar{x} \approx \frac{\text{system moment}}{\text{system mass}} \approx \frac{\sum x_k \Delta m_k}{\sum \Delta m_k} \approx \frac{\sum x_k \delta(x_k) \Delta x_k}{\sum \delta(x_k) \Delta x_k}. \quad (2)$$

The sum in the last numerator in Equation (2) is a Riemann sum for the continuous function $x\delta(x)$ over the closed interval $[a, b]$. The sum in the denominator is a Riemann sum for the function $\delta(x)$ over this interval. We expect the approximations in Equation (2) to improve as the strip is partitioned more finely, and we are led to the equation

$$\bar{x} = \frac{\int_a^b x\delta(x) dx}{\int_a^b \delta(x) dx}.$$

This is the formula we use to find $\bar{x}$.

Moment, Mass, and Center of Mass of a Thin Rod or Strip Along the x -Axis with Density Function $\delta(x)$

$$\text{Moment about the origin:} \quad M_0 = \int_a^b x\delta(x) dx \quad (3a)$$

$$\text{Mass:} \quad M = \int_a^b \delta(x) dx \quad (3b)$$

$$\text{Center of mass:} \quad \bar{x} = \frac{M_0}{M} \quad (3c)$$

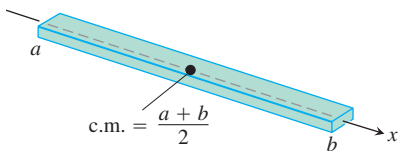


FIGURE 6.30 The center of mass of a straight, thin rod or strip of constant density lies halfway between its ends (Example 1).

EXAMPLE 1 Strips and Rods of Constant Density

Show that the center of mass of a straight, thin strip or rod of constant density lies halfway between its two ends.

Solution We model the strip as a portion of the x -axis from $x = a$ to $x = b$ (Figure 6.30). Our goal is to show that $\bar{x} = (a + b)/2$, the point halfway between a and b .

The key is the density's having a constant value. This enables us to regard the function $\delta(x)$ in the integrals in Equation (3) as a constant (call it δ), with the result that

$$M_0 = \int_a^b \delta x \, dx = \delta \int_a^b x \, dx = \delta \left[\frac{1}{2} x^2 \right]_a^b = \frac{\delta}{2} (b^2 - a^2)$$

$$M = \int_a^b \delta \, dx = \delta \int_a^b 1 \, dx = \delta [x]_a^b = \delta(b - a)$$

$$\begin{aligned} \bar{x} &= \frac{M_0}{M} = \frac{\frac{\delta}{2} (b^2 - a^2)}{\delta(b - a)} \\ &= \frac{a + b}{2}. \end{aligned}$$

The δ 's cancel in the formula for $\bar{x}$.

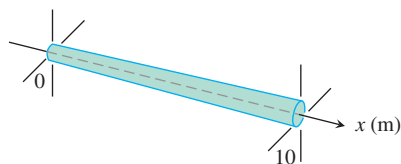


FIGURE 6.31 We can treat a rod of variable thickness as a rod of variable density (Example 2).

EXAMPLE 2 Variable-Density Rod

The 10-m-long rod in Figure 6.31 thickens from left to right so that its density, instead of being constant, is $\delta(x) = 1 + (x/10)$ kg/m. Find the rod's center of mass.

Solution The rod's moment about the origin (Equation 3a) is

$$\begin{aligned} M_0 &= \int_0^{10} x\delta(x) \, dx = \int_0^{10} x \left(1 + \frac{x}{10} \right) dx = \int_0^{10} \left(x + \frac{x^2}{10} \right) dx \\ &= \left[\frac{x^2}{2} + \frac{x^3}{30} \right]_0^{10} = 50 + \frac{100}{3} = \frac{250}{3} \text{ kg} \cdot \text{m}. \end{aligned}$$

The units of a moment are mass $\times$ length.

The rod's mass (Equation 3b) is

$$M = \int_0^{10} \delta(x) \, dx = \int_0^{10} \left(1 + \frac{x}{10} \right) dx = \left[x + \frac{x^2}{20} \right]_0^{10} = 10 + 5 = 15 \text{ kg}.$$

The center of mass (Equation 3c) is located at the point

$$\bar{x} = \frac{M_0}{M} = \frac{250}{3} \cdot \frac{1}{15} = \frac{50}{9} \approx 5.56 \text{ m}.$$

Masses Distributed over a Plane Region

Suppose that we have a finite collection of masses located in the plane, with mass m_k at the point (x_k, y_k) (see Figure 6.32). The mass of the system is

$$\text{System mass: } M = \sum m_k.$$

Each mass m_k has a moment about each axis. Its moment about the x -axis is $m_k y_k$, and its moment about the y -axis is $m_k x_k$. The moments of the entire system about the two axes are

$$\text{Moment about } x\text{-axis: } M_x = \sum m_k y_k,$$

$$\text{Moment about } y\text{-axis: } M_y = \sum m_k x_k.$$

The x -coordinate of the system's center of mass is defined to be

$$\bar{x} = \frac{M_y}{M} = \frac{\sum m_k x_k}{\sum m_k}. \quad (4)$$

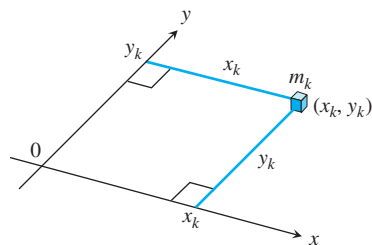


FIGURE 6.32 Each mass m_k has a moment about each axis.

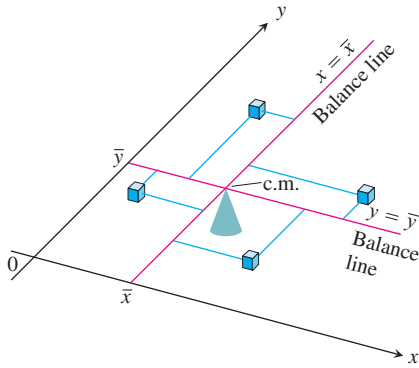


FIGURE 6.33 A two-dimensional array of masses balances on its center of mass.

With this choice of $\bar{x}$, as in the one-dimensional case, the system balances about the line $x = \bar{x}$ (Figure 6.33).

The y -coordinate of the system's center of mass is defined to be

$$\bar{y} = \frac{M_x}{M} = \frac{\sum m_k y_k}{\sum m_k}. \quad (5)$$

With this choice of $\bar{y}$, the system balances about the line $y = \bar{y}$ as well. The torques exerted by the masses about the line $y = \bar{y}$ cancel out. Thus, as far as balance is concerned, the system behaves as if all its mass were at the single point $(\bar{x}, \bar{y})$. We call this point the system's **center of mass**.

Thin, Flat Plates

In many applications, we need to find the center of mass of a thin, flat plate: a disk of aluminum, say, or a triangular sheet of steel. In such cases, we assume the distribution of mass to be continuous, and the formulas we use to calculate $\bar{x}$ and $\bar{y}$ contain integrals instead of finite sums. The integrals arise in the following way.

Imagine the plate occupying a region in the xy -plane, cut into thin strips parallel to one of the axes (in Figure 6.34, the y -axis). The center of mass of a typical strip is $(\tilde{x}, \tilde{y})$. We treat the strip's mass Δm as if it were concentrated at $(\tilde{x}, \tilde{y})$. The moment of the strip about the y -axis is then $\tilde{x} \Delta m$. The moment of the strip about the x -axis is $\tilde{y} \Delta m$. Equations (4) and (5) then become

$$\bar{x} = \frac{M_y}{M} = \frac{\sum \tilde{x} \Delta m}{\sum \Delta m}, \quad \bar{y} = \frac{M_x}{M} = \frac{\sum \tilde{y} \Delta m}{\sum \Delta m}.$$

As in the one-dimensional case, the sums are Riemann sums for integrals and approach these integrals as limiting values as the strips into which the plate is cut become narrower and narrower. We write these integrals symbolically as

$$\bar{x} = \frac{\int \tilde{x} \, dm}{\int dm} \quad \text{and} \quad \bar{y} = \frac{\int \tilde{y} \, dm}{\int dm}.$$

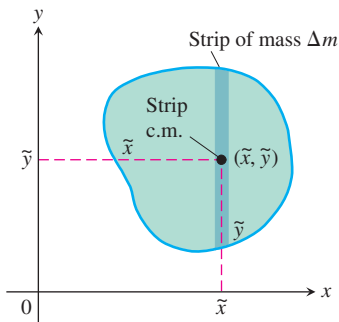


FIGURE 6.34 A plate cut into thin strips parallel to the y -axis. The moment exerted by a typical strip about each axis is the moment its mass Δm would exert if concentrated at the strip's center of mass $(\tilde{x}, \tilde{y})$.

Moments, Mass, and Center of Mass of a Thin Plate Covering a Region in the xy -Plane

$$\begin{aligned} \text{Moment about the } x\text{-axis:} \quad M_x &= \int \tilde{y} \, dm \\ \text{Moment about the } y\text{-axis:} \quad M_y &= \int \tilde{x} \, dm \\ \text{Mass:} \quad M &= \int dm \\ \text{Center of mass:} \quad \bar{x} &= \frac{M_y}{M}, \quad \bar{y} = \frac{M_x}{M} \end{aligned} \quad (6)$$

To evaluate these integrals, we picture the plate in the coordinate plane and sketch a strip of mass parallel to one of the coordinate axes. We then express the strip's mass dm and the coordinates $(\tilde{x}, \tilde{y})$ of the strip's center of mass in terms of x or y . Finally, we integrate $\tilde{y} \, dm$, $\tilde{x} \, dm$, and dm between limits of integration determined by the plate's location in the plane.

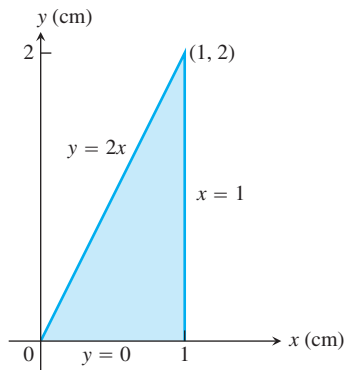


FIGURE 6.35 The plate in Example 3.

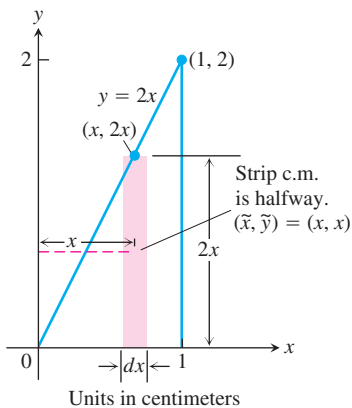


FIGURE 6.36 Modeling the plate in Example 3 with vertical strips.

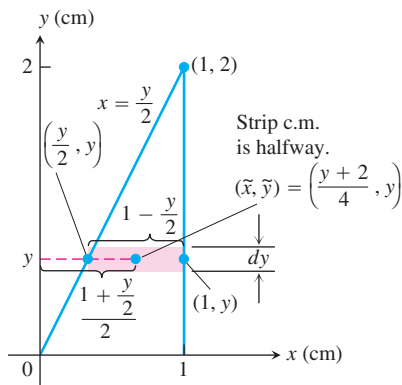


FIGURE 6.37 Modeling the plate in Example 3 with horizontal strips.

EXAMPLE 3 Constant-Density Plate

The triangular plate shown in Figure 6.35 has a constant density of $\delta = 3 \text{ g/cm}^2$. Find

- the plate's moment M_y about the y -axis.
- the plate's mass M .
- the x -coordinate of the plate's center of mass (c.m.).

Solution

Method 1: Vertical Strips (Figure 6.36)

- (a) The moment M_y : The typical vertical strip has

$$\text{center of mass (c.m.): } (\tilde{x}, \tilde{y}) = (x, x)$$

$$\text{length: } 2x$$

$$\text{width: } dx$$

$$\text{area: } dA = 2x \, dx$$

$$\text{mass: } dm = \delta \, dA = 3 \cdot 2x \, dx = 6x \, dx$$

$$\text{distance of c.m. from } y\text{-axis: } \tilde{x} = x.$$

The moment of the strip about the y -axis is

$$\tilde{x} \, dm = x \cdot 6x \, dx = 6x^2 \, dx.$$

The moment of the plate about the y -axis is therefore

$$M_y = \int \tilde{x} \, dm = \int_0^1 6x^2 \, dx = 2x^3 \Big|_0^1 = 2 \text{ g} \cdot \text{cm}.$$

- (b) The plate's mass:

$$M = \int dm = \int_0^1 6x \, dx = 3x^2 \Big|_0^1 = 3 \text{ g}.$$

- (c) The x -coordinate of the plate's center of mass:

$$\bar{x} = \frac{M_y}{M} = \frac{2 \text{ g} \cdot \text{cm}}{3 \text{ g}} = \frac{2}{3} \text{ cm}.$$

By a similar computation, we could find M_x and $\bar{y} = M_x/M$.

Method 2: Horizontal Strips (Figure 6.37)

- (a) The moment M_y : The y -coordinate of the center of mass of a typical horizontal strip is y (see the figure), so

$$\tilde{y} = y.$$

The x -coordinate is the x -coordinate of the point halfway across the triangle. This makes it the average of $y/2$ (the strip's left-hand x -value) and 1 (the strip's right-hand x -value):

$$\tilde{x} = \frac{(y/2) + 1}{2} = \frac{y}{4} + \frac{1}{2} = \frac{y + 2}{4}.$$

We also have

$$\text{length: } 1 - \frac{y}{2} = \frac{2-y}{2}$$

$$\text{width: } dy$$

$$\text{area: } dA = \frac{2-y}{2} dy$$

$$\text{mass: } dm = \delta dA = 3 \cdot \frac{2-y}{2} dy$$

$$\text{distance of c.m. to } y\text{-axis: } \tilde{x} = \frac{y+2}{4}.$$

The moment of the strip about the y -axis is

$$\tilde{x} dm = \frac{y+2}{4} \cdot 3 \cdot \frac{2-y}{2} dy = \frac{3}{8} (4-y^2) dy.$$

The moment of the plate about the y -axis is

$$M_y = \int \tilde{x} dm = \int_0^2 \frac{3}{8} (4-y^2) dy = \frac{3}{8} \left[4y - \frac{y^3}{3} \right]_0^2 = \frac{3}{8} \left(\frac{16}{3} \right) = 2 \text{ g} \cdot \text{cm}.$$

(b) The plate's mass:

$$M = \int dm = \int_0^2 \frac{3}{2} (2-y) dy = \frac{3}{2} \left[2y - \frac{y^2}{2} \right]_0^2 = \frac{3}{2} (4-2) = 3 \text{ g}.$$

(c) The x -coordinate of the plate's center of mass:

$$\bar{x} = \frac{M_y}{M} = \frac{2 \text{ g} \cdot \text{cm}}{3 \text{ g}} = \frac{2}{3} \text{ cm}.$$

By a similar computation, we could find M_x and $\bar{y}$. ■

If the distribution of mass in a thin, flat plate has an axis of symmetry, the center of mass will lie on this axis. If there are two axes of symmetry, the center of mass will lie at their intersection. These facts often help to simplify our work.

EXAMPLE 4 Constant-Density Plate

Find the center of mass of a thin plate of constant density δ covering the region bounded above by the parabola $y = 4 - x^2$ and below by the x -axis (Figure 6.38).

Solution Since the plate is symmetric about the y -axis and its density is constant, the distribution of mass is symmetric about the y -axis and the center of mass lies on the y -axis. Thus, $\bar{x} = 0$. It remains to find $\bar{y} = M_x/M$.

A trial calculation with horizontal strips (Figure 6.38a) leads to an inconvenient integration

$$M_x = \int_0^4 2\delta y \sqrt{4-y} dy.$$

We therefore model the distribution of mass with vertical strips instead (Figure 6.38b).

How to Find a Plate's Center of Mass

1. Picture the plate in the xy -plane.
2. Sketch a strip of mass parallel to one of the coordinate axes and find its dimensions.
3. Find the strip's mass dm and center of mass $(\tilde{x}, \tilde{y})$.
4. Integrate $\tilde{y} dm$, $\tilde{x} dm$, and dm to find M_x , M_y , and M .
5. Divide the moments by the mass to calculate $\bar{x}$ and $\bar{y}$.

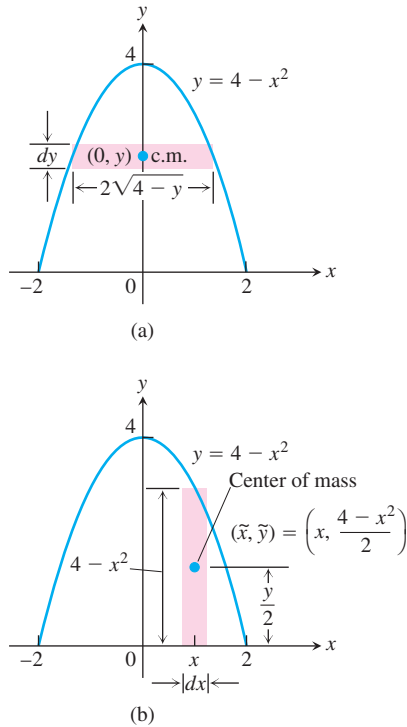


FIGURE 6.38 Modeling the plate in Example 4 with (a) horizontal strips leads to an inconvenient integration, so we model with (b) vertical strips instead.

The typical vertical strip has

$$\text{center of mass (c.m.):} \quad (\tilde{x}, \tilde{y}) = \left(x, \frac{4 - x^2}{2}\right)$$

$$\text{length:} \quad 4 - x^2$$

$$\text{width:} \quad dx$$

$$\text{area:} \quad dA = (4 - x^2) dx$$

$$\text{mass:} \quad dm = \delta dA = \delta(4 - x^2) dx$$

$$\text{distance from c.m. to } x\text{-axis:} \quad \tilde{y} = \frac{4 - x^2}{2}.$$

The moment of the strip about the x -axis is

$$\tilde{y} dm = \frac{4 - x^2}{2} \cdot \delta(4 - x^2) dx = \frac{\delta}{2} (4 - x^2)^2 dx.$$

The moment of the plate about the x -axis is

$$\begin{aligned} M_x &= \int \tilde{y} dm = \int_{-2}^2 \frac{\delta}{2} (4 - x^2)^2 dx \\ &= \frac{\delta}{2} \int_{-2}^2 (16 - 8x^2 + x^4) dx = \frac{256}{15} \delta. \end{aligned} \quad (7)$$

The mass of the plate is

$$M = \int dm = \int_{-2}^2 \delta(4 - x^2) dx = \frac{32}{3} \delta. \quad (8)$$

Therefore,

$$\bar{y} = \frac{M_x}{M} = \frac{(256/15) \delta}{(32/3) \delta} = \frac{8}{5}.$$

The plate's center of mass is the point

$$(\bar{x}, \bar{y}) = \left(0, \frac{8}{5}\right).$$

EXAMPLE 5 Variable-Density Plate

Find the center of mass of the plate in Example 4 if the density at the point (x, y) is $\delta = 2x^2$, twice the square of the distance from the point to the y -axis.

Solution The mass distribution is still symmetric about the y -axis, so $\bar{x} = 0$. With $\delta = 2x^2$, Equations (7) and (8) become

$$\begin{aligned} M_x &= \int \tilde{y} \, dm = \int_{-2}^2 \frac{\delta}{2} (4 - x^2)^2 \, dx = \int_{-2}^2 x^2 (4 - x^2)^2 \, dx \\ &= \int_{-2}^2 (16x^2 - 8x^4 + x^6) \, dx = \frac{2048}{105} \end{aligned} \quad (7')$$

$$\begin{aligned} M &= \int dm = \int_{-2}^2 \delta(4 - x^2) \, dx = \int_{-2}^2 2x^2(4 - x^2) \, dx \\ &= \int_{-2}^2 (8x^2 - 2x^4) \, dx = \frac{256}{15}. \end{aligned} \quad (8')$$

Therefore,

$$\bar{y} = \frac{M_x}{M} = \frac{2048}{105} \cdot \frac{15}{256} = \frac{8}{7}.$$

The plate's new center of mass is

$$(\bar{x}, \bar{y}) = \left(0, \frac{8}{7}\right).$$

EXAMPLE 6 Constant-Density Wire

Find the center of mass of a wire of constant density δ shaped like a semicircle of radius a .

Solution We model the wire with the semicircle $y = \sqrt{a^2 - x^2}$ (Figure 6.39). The distribution of mass is symmetric about the y -axis, so $\bar{x} = 0$. To find $\bar{y}$, we imagine the wire divided into short segments. The typical segment (Figure 6.39a) has

$$\text{length: } ds = a \, d\theta$$

$$\text{mass: } dm = \delta \, ds = \delta a \, d\theta$$

$$\text{distance of c.m. to } x\text{-axis: } \tilde{y} = a \sin \theta.$$

Mass per unit length
times length

Hence,

$$\bar{y} = \frac{\int \tilde{y} \, dm}{\int dm} = \frac{\int_0^\pi a \sin \theta \cdot \delta a \, d\theta}{\int_0^\pi \delta a \, d\theta} = \frac{\delta a^2 [-\cos \theta]_0^\pi}{\delta a \pi} = \frac{2}{\pi} a.$$

The center of mass lies on the axis of symmetry at the point $(0, 2a/\pi)$, about two-thirds of the way up from the origin (Figure 6.39b).

Centroids

When the density function is constant, it cancels out of the numerator and denominator of the formulas for $\bar{x}$ and $\bar{y}$. This happened in nearly every example in this section. As far as $\bar{x}$ and $\bar{y}$ were concerned, δ might as well have been 1. Thus, when the density is constant, the location of the center of mass is a feature of the geometry of the object and not of the material from which it is made. In such cases, engineers may call the center of mass the **centroid** of the shape, as in “Find the centroid of a triangle or a solid cone.” To do so, just set δ equal to 1 and proceed to find $\bar{x}$ and $\bar{y}$ as before, by dividing moments by masses.

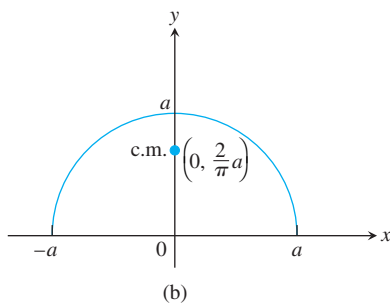
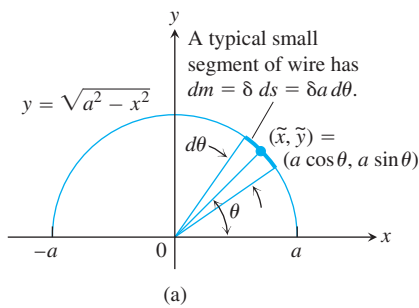
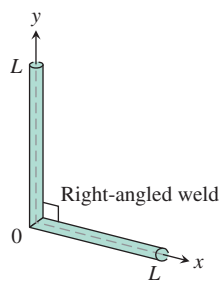


FIGURE 6.39 The semicircular wire in Example 6. (a) The dimensions and variables used in finding the center of mass. (b) The center of mass does not lie on the wire.

EXERCISES 6.4

Thin Rods

1. An 80-lb child and a 100-lb child are balancing on a seesaw. The 80-lb child is 5 ft from the fulcrum. How far from the fulcrum is the 100-lb child?
2. The ends of a log are placed on two scales. One scale reads 100 kg and the other 200 kg. Where is the log's center of mass?
3. The ends of two thin steel rods of equal length are welded together to make a right-angled frame. Locate the frame's center of mass. (*Hint*: Where is the center of mass of each rod?)



4. You weld the ends of two steel rods into a right-angled frame. One rod is twice the length of the other. Where is the frame's center of mass? (*Hint*: Where is the center of mass of each rod?)

Exercises 5–12 give density functions of thin rods lying along various intervals of the x -axis. Use Equations (3a) through (3c) to find each rod's moment about the origin, mass, and center of mass.

5. $\delta(x) = 4$, $0 \leq x \leq 2$
6. $\delta(x) = 4$, $1 \leq x \leq 3$
7. $\delta(x) = 1 + (x/3)$, $0 \leq x \leq 3$
8. $\delta(x) = 2 - (x/4)$, $0 \leq x \leq 4$
9. $\delta(x) = 1 + (1/\sqrt{x})$, $1 \leq x \leq 4$
10. $\delta(x) = 3(x^{-3/2} + x^{-5/2})$, $0.25 \leq x \leq 1$
11. $\delta(x) = \begin{cases} 2 - x, & 0 \leq x < 1 \\ x, & 1 \leq x \leq 2 \end{cases}$
12. $\delta(x) = \begin{cases} x + 1, & 0 \leq x < 1 \\ 2, & 1 \leq x \leq 2 \end{cases}$

Thin Plates with Constant Density

In Exercises 13–24, find the center of mass of a thin plate of constant density δ covering the given region.

13. The region bounded by the parabola $y = x^2$ and the line $y = 4$
14. The region bounded by the parabola $y = 25 - x^2$ and the x -axis
15. The region bounded by the parabola $y = x - x^2$ and the line $y = -x$
16. The region enclosed by the parabolas $y = x^2 - 3$ and $y = -2x^2$

17. The region bounded by the y -axis and the curve $x = y - y^3$, $0 \leq y \leq 1$
18. The region bounded by the parabola $x = y^2 - y$ and the line $y = x$
19. The region bounded by the x -axis and the curve $y = \cos x$, $-\pi/2 \leq x \leq \pi/2$
20. The region between the x -axis and the curve $y = \sec^2 x$, $-\pi/4 \leq x \leq \pi/4$
21. The region bounded by the parabolas $y = 2x^2 - 4x$ and $y = 2x - x^2$
22. a. The region cut from the first quadrant by the circle $x^2 + y^2 = 9$
b. The region bounded by the x -axis and the semicircle $y = \sqrt{9 - x^2}$
Compare your answer in part (b) with the answer in part (a).
23. The “triangular” region in the first quadrant between the circle $x^2 + y^2 = 9$ and the lines $x = 3$ and $y = 3$. (*Hint*: Use geometry to find the area.)
24. The region bounded above by the curve $y = 1/x^3$, below by the curve $y = -1/x^3$, and on the left and right by the lines $x = 1$ and $x = a > 1$. Also, find $\lim_{a \rightarrow \infty} \bar{x}$.

Thin Plates with Varying Density

25. Find the center of mass of a thin plate covering the region between the x -axis and the curve $y = 2/x^2$, $1 \leq x \leq 2$, if the plate's density at the point (x, y) is $\delta(x) = x^2$.
26. Find the center of mass of a thin plate covering the region bounded below by the parabola $y = x^2$ and above by the line $y = x$ if the plate's density at the point (x, y) is $\delta(x) = 12x$.
27. The region bounded by the curves $y = \pm 4/\sqrt{x}$ and the lines $x = 1$ and $x = 4$ is revolved about the y -axis to generate a solid.
a. Find the volume of the solid.
b. Find the center of mass of a thin plate covering the region if the plate's density at the point (x, y) is $\delta(x) = 1/x$.
c. Sketch the plate and show the center of mass in your sketch.
28. The region between the curve $y = 2/x$ and the x -axis from $x = 1$ to $x = 4$ is revolved about the x -axis to generate a solid.
a. Find the volume of the solid.
b. Find the center of mass of a thin plate covering the region if the plate's density at the point (x, y) is $\delta(x) = \sqrt{x}$.
c. Sketch the plate and show the center of mass in your sketch.

Centroids of Triangles

29. The centroid of a triangle lies at the intersection of the triangle's medians (*Figure 6.40a*) You may recall that the point

inside a triangle that lies one-third of the way from each side toward the opposite vertex is the point where the triangle's three medians intersect. Show that the centroid lies at the intersection of the medians by showing that it too lies one-third of the way from each side toward the opposite vertex. To do so, take the following steps.

- i. Stand one side of the triangle on the x -axis as in Figure 6.40b. Express dm in terms of L and dy .
- ii. Use similar triangles to show that $L = (b/h)(h - y)$. Substitute this expression for L in your formula for dm .
- iii. Show that $\bar{y} = h/3$.
- iv. Extend the argument to the other sides.

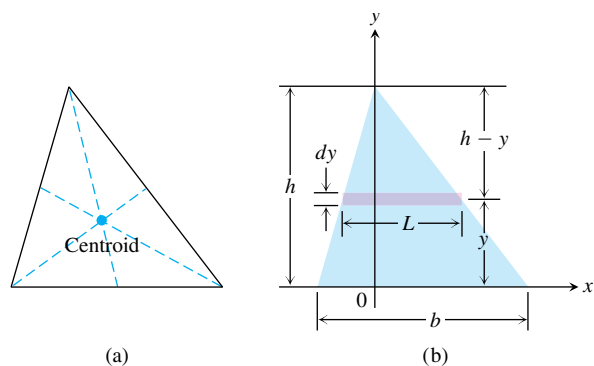


FIGURE 6.40 The triangle in Exercise 29. (a) The centroid. (b) The dimensions and variables to use in locating the center of mass.

Use the result in Exercise 29 to find the centroids of the triangles whose vertices appear in Exercises 30–34. Assume $a, b > 0$.

30. $(-1, 0), (1, 0), (0, 3)$
31. $(0, 0), (1, 0), (0, 1)$
32. $(0, 0), (a, 0), (0, a)$
33. $(0, 0), (a, 0), (0, b)$
34. $(0, 0), (a, 0), (a/2, b)$

Thin Wires

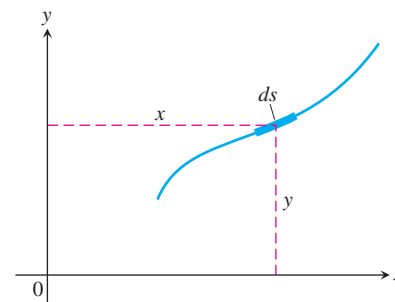
35. **Constant density** Find the moment about the x -axis of a wire of constant density that lies along the curve $y = \sqrt{x}$ from $x = 0$ to $x = 2$.
36. **Constant density** Find the moment about the x -axis of a wire of constant density that lies along the curve $y = x^3$ from $x = 0$ to $x = 1$.
37. **Variable density** Suppose that the density of the wire in Example 6 is $\delta = k \sin \theta$ (k constant). Find the center of mass.
38. **Variable density** Suppose that the density of the wire in Example 6 is $\delta = 1 + k|\cos \theta|$ (k constant). Find the center of mass.

Engineering Formulas

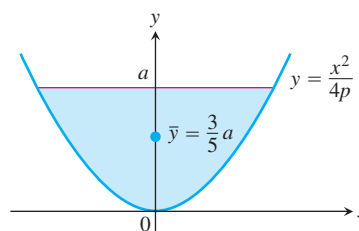
Verify the statements and formulas in Exercises 39–42.

39. The coordinates of the centroid of a differentiable plane curve are

$$\bar{x} = \frac{\int x \, ds}{\text{length}}, \quad \bar{y} = \frac{\int y \, ds}{\text{length}}.$$

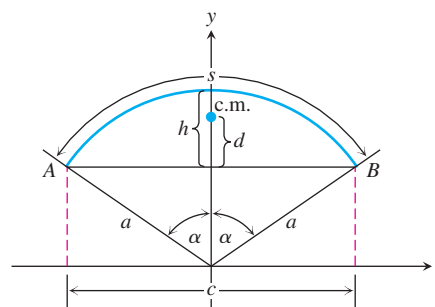


40. Whatever the value of $p > 0$ in the equation $y = x^2/(4p)$, the y -coordinate of the centroid of the parabolic segment shown here is $\bar{y} = (3/5)a$.



41. For wires and thin rods of constant density shaped like circular arcs centered at the origin and symmetric about the y -axis, the y -coordinate of the center of mass is

$$\bar{y} = \frac{a \sin \alpha}{\alpha} = \frac{ac}{s}.$$



42. (Continuation of Exercise 41.)

- a. Show that when α is small, the distance d from the centroid to chord AB is about $2h/3$ (in the notation of the figure here) by taking the following steps.

- i. Show that

$$\frac{d}{h} = \frac{\sin \alpha - \alpha \cos \alpha}{\alpha - \alpha \cos \alpha}. \quad (9)$$

- ii. Graph

$$f(\alpha) = \frac{\sin \alpha - \alpha \cos \alpha}{\alpha - \alpha \cos \alpha}$$

and use the trace feature to show that $\lim_{\alpha \rightarrow 0^+} f(\alpha) \approx 2/3$.

- b. The error (difference between d and $2h/3$) is small even for angles greater than 45° . See for yourself by evaluating the right-hand side of Equation (9) for $\alpha = 0.2, 0.4, 0.6, 0.8$, and 1.0 rad.

6.5

Areas of Surfaces of Revolution and the Theorems of Pappus

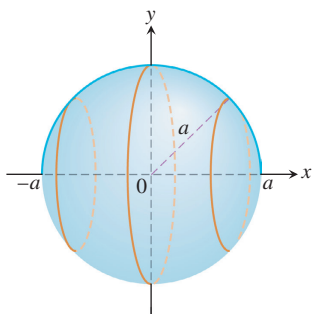


FIGURE 6.41 Rotating the semicircle $y = \sqrt{a^2 - x^2}$ of radius a with center at the origin generates a spherical surface with area $4\pi a^2$.

When you jump rope, the rope sweeps out a surface in the space around you called a *surface of revolution*. The “area” of this surface depends on the length of the rope and the distance of each of its segments from the axis of revolution. In this section we define areas of surfaces of revolution. More complicated surfaces are treated in Chapter 16.

Defining Surface Area

We want our definition of the area of a surface of revolution to be consistent with known results from classical geometry for the surface areas of spheres, circular cylinders, and cones. So if the jump rope discussed in the introduction takes the shape of a semicircle with radius a rotated about the x -axis (Figure 6.41), it generates a sphere with surface area $4\pi a^2$.

Before considering general curves, we begin by rotating horizontal and slanted line segments about the x -axis. If we rotate the horizontal line segment AB having length Δx about the x -axis (Figure 6.42a), we generate a cylinder with surface area $2\pi y \Delta x$. This area is the same as that of a rectangle with side lengths Δx and $2\pi y$ (Figure 6.42b). The length $2\pi y$ is the circumference of the circle of radius y generated by rotating the point (x, y) on the line AB about the x -axis.

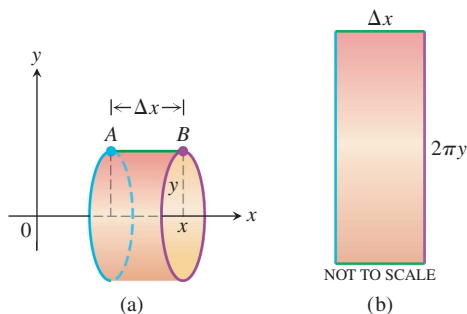


FIGURE 6.42 (a) A cylindrical surface generated by rotating the horizontal line segment AB of length Δx about the x -axis has area $2\pi y \Delta x$. (b) The cut and rolled out cylindrical surface as a rectangle.

Suppose the line segment AB has length Δs and is slanted rather than horizontal. Now when AB is rotated about the x -axis, it generates a frustum of a cone (Figure 6.43a). From classical geometry, the surface area of this frustum is $2\pi y^* \Delta s$, where $y^* = (y_1 + y_2)/2$ is the average height of the slanted segment AB above the x -axis. This surface area is the same as that of a rectangle with side lengths Δs and $2\pi y^*$ (Figure 6.43b).

Let's build on these geometric principles to define the area of a surface swept out by revolving more general curves about the x -axis. Suppose we want to find the area of the surface swept out by revolving the graph of a nonnegative continuous function $y = f(x)$, $a \leq x \leq b$, about the x -axis. We partition the closed interval $[a, b]$ in the usual way and use the points in the partition to subdivide the graph into short arcs. Figure 6.44 shows a typical arc PQ and the band it sweeps out as part of the graph of f .

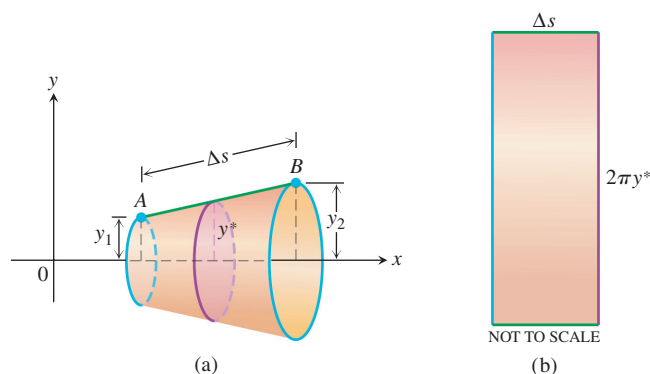


FIGURE 6.43 (a) The frustum of a cone generated by rotating the slanted line segment AB of length Δs about the x -axis has area $2\pi y^* \Delta s$. (b) The area of the rectangle for $y^* = \frac{y_1 + y_2}{2}$, the average height of AB above the x -axis.

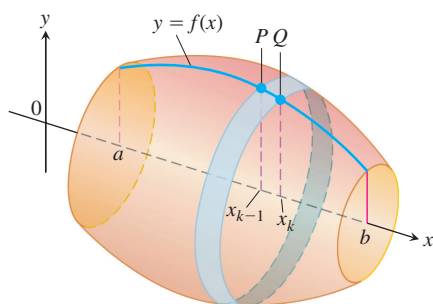


FIGURE 6.44 The surface generated by revolving the graph of a nonnegative function $y = f(x)$, $a \leq x \leq b$, about the x -axis. The surface is a union of bands like the one swept out by the arc PQ .

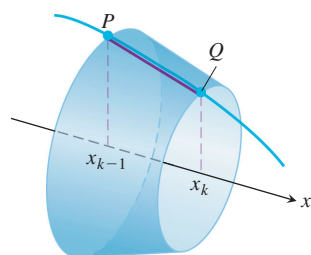


FIGURE 6.45 The line segment joining P and Q sweeps out a frustum of a cone.

As the arc PQ revolves about the x -axis, the line segment joining P and Q sweeps out a frustum of a cone whose axis lies along the x -axis (Figure 6.45). The surface area of this frustum approximates the surface area of the band swept out by the arc PQ . The surface area of the frustum of the cone shown in Figure 6.45 is $2\pi y^* L$, where y^* is the average height of the line segment joining P and Q , and L is its length (just as before). Since $f \geq 0$, from Figure 6.46 we see that the average height of the line segment is $y^* = (f(x_{k-1}) + f(x_k))/2$, and the slant length is $L = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$. Therefore,

$$\begin{aligned} \text{Frustum surface area} &= 2\pi \cdot \frac{f(x_{k-1}) + f(x_k)}{2} \cdot \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} \\ &= \pi(f(x_{k-1}) + f(x_k))\sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}. \end{aligned}$$

The area of the original surface, being the sum of the areas of the bands swept out by arcs like arc PQ , is approximated by the frustum area sum

$$\sum_{k=1}^n \pi(f(x_{k-1}) + f(x_k))\sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}. \quad (1)$$

We expect the approximation to improve as the partition of $[a, b]$ becomes finer. Moreover, if the function f is differentiable, then by the Mean Value Theorem, there is a point $(c_k, f(c_k))$ on the curve between P and Q where the tangent is parallel to the segment PQ (Figure 6.47). At this point,

$$\begin{aligned} f'(c_k) &= \frac{\Delta y_k}{\Delta x_k}, \\ \Delta y_k &= f'(c_k) \Delta x_k. \end{aligned}$$

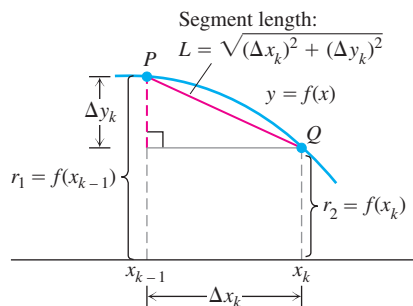


FIGURE 6.46 Dimensions associated with the arc and line segment PQ .

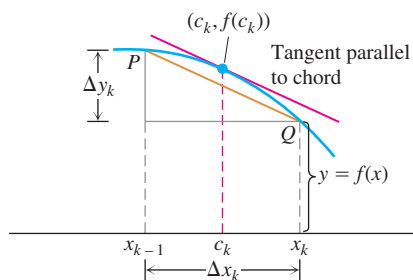


FIGURE 6.47 If f is smooth, the Mean Value Theorem guarantees the existence of a point c_k where the tangent is parallel to segment PQ .

With this substitution for Δy_k , the sums in Equation (1) take the form

$$\begin{aligned} \sum_{k=1}^n \pi(f(x_{k-1}) + f(x_k)) \sqrt{(\Delta x_k)^2 + (f'(c_k) \Delta x_k)^2} \\ = \sum_{k=1}^n \pi(f(x_{k-1}) + f(x_k)) \sqrt{1 + (f'(c_k))^2} \Delta x_k. \end{aligned} \quad (2)$$

These sums are not the Riemann sums of any function because the points x_{k-1} , x_k , and c_k are not the same. However, a theorem from advanced calculus assures us that as the norm of the partition of $[a, b]$ goes to zero, the sums in Equation (2) converge to the integral

$$\int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx.$$

We therefore define this integral to be the area of the surface swept out by the graph of f from a to b .

DEFINITION Surface Area for Revolution About the x -Axis

If the function $f(x) \geq 0$ is continuously differentiable on $[a, b]$, the **area** of the surface generated by revolving the curve $y = f(x)$ about the x -axis is

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx. \quad (3)$$

The square root in Equation (3) is the same one that appears in the formula for the length of the generating curve in Equation (2) of Section 6.3.

EXAMPLE 1 Applying the Surface Area Formula

Find the area of the surface generated by revolving the curve $y = 2\sqrt{x}$, $1 \leq x \leq 2$, about the x -axis (Figure 6.48).

Solution We evaluate the formula

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{Eq. (3)}$$

with

$$\begin{aligned} a = 1, \quad b = 2, \quad y = 2\sqrt{x}, \quad \frac{dy}{dx} = \frac{1}{\sqrt{x}}, \\ \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \left(\frac{1}{\sqrt{x}}\right)^2} \\ = \sqrt{1 + \frac{1}{x}} = \sqrt{\frac{x+1}{x}} = \frac{\sqrt{x+1}}{\sqrt{x}}. \end{aligned}$$

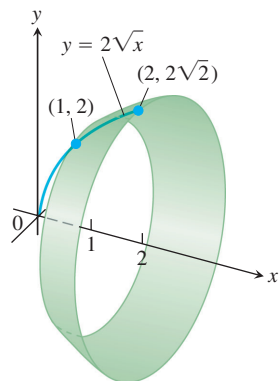


FIGURE 6.48 In Example 1 we calculate the area of this surface.

With these substitutions,

$$\begin{aligned} S &= \int_1^2 2\pi \cdot 2\sqrt{x} \frac{\sqrt{x+1}}{\sqrt{x}} dx = 4\pi \int_1^2 \sqrt{x+1} dx \\ &= 4\pi \cdot \frac{2}{3} (x+1)^{3/2} \Big|_1^2 = \frac{8\pi}{3} (3\sqrt{3} - 2\sqrt{2}). \end{aligned}$$

Revolution About the y-Axis

For revolution about the y-axis, we interchange x and y in Equation (3).

Surface Area for Revolution About the y-Axis

If $x = g(y) \geq 0$ is continuously differentiable on $[c, d]$, the area of the surface generated by revolving the curve $x = g(y)$ about the y-axis is

$$S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_c^d 2\pi g(y) \sqrt{1 + (g'(y))^2} dy. \quad (4)$$

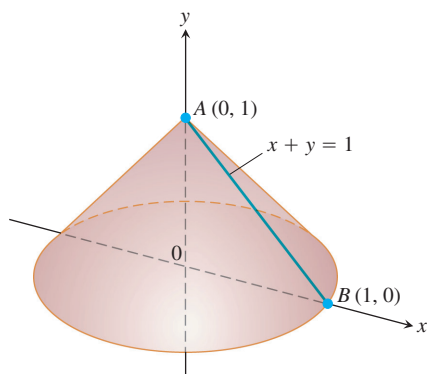


FIGURE 6.49 Revolving line segment AB about the y -axis generates a cone whose lateral surface area we can now calculate in two different ways (Example 2).

EXAMPLE 2 Finding Area for Revolution about the y-Axis

The line segment $x = 1 - y$, $0 \leq y \leq 1$, is revolved about the y -axis to generate the cone in Figure 6.49. Find its lateral surface area (which excludes the base area).

Solution Here we have a calculation we can check with a formula from geometry:

$$\text{Lateral surface area} = \frac{\text{base circumference}}{2} \times \text{slant height} = \pi\sqrt{2}.$$

To see how Equation (4) gives the same result, we take

$$\begin{aligned} c &= 0, \quad d = 1, \quad x = 1 - y, \quad \frac{dx}{dy} = -1, \\ \sqrt{1 + \left(\frac{dx}{dy}\right)^2} &= \sqrt{1 + (-1)^2} = \sqrt{2} \end{aligned}$$

and calculate

$$\begin{aligned} S &= \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_0^1 2\pi(1 - y)\sqrt{2} dy \\ &= 2\pi\sqrt{2} \left[y - \frac{y^2}{2} \right]_0^1 = 2\pi\sqrt{2} \left(1 - \frac{1}{2} \right) \\ &= \pi\sqrt{2}. \end{aligned}$$

The results agree, as they should.

Parametrized Curves

Regardless of the coordinate axis of revolution, the square roots appearing in Equations (3) and (4) are the same ones that appear in the formulas for arc length in Section 6.3. If the curve is parametrized by the equations $x = f(t)$ and $y = g(t)$, $a \leq t \leq b$, where f and g are continuously differentiable on $[a, b]$, then the corresponding square root appearing in the arc length formula is

$$\sqrt{[f'(t)]^2 + [g'(t)]^2} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}.$$

This observation leads to the following formulas for area of surfaces of revolution for smooth parametrized curves.

Surface Area of Revolution for Parametrized Curves

If a smooth curve $x = f(t)$, $y = g(t)$, $a \leq t \leq b$, is traversed exactly once as t increases from a to b , then the areas of the surfaces generated by revolving the curve about the coordinate axes are as follows.

1. Revolution about the x -axis ($y \geq 0$):

$$S = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \quad (5)$$

2. Revolution about the y -axis ($x \geq 0$):

$$S = \int_a^b 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \quad (6)$$

As with length, we can calculate surface area from any convenient parametrization that meets the stated criteria.

EXAMPLE 3 Applying Surface Area Formula

The standard parametrization of the circle of radius 1 centered at the point $(0, 1)$ in the xy -plane is

$$x = \cos t, \quad y = 1 + \sin t, \quad 0 \leq t \leq 2\pi.$$

Use this parametrization to find the area of the surface swept out by revolving the circle about the x -axis (Figure 6.50).

Solution We evaluate the formula

$$\begin{aligned} S &= \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt && \text{Eq. (5) for revolution} \\ &= \int_0^{2\pi} 2\pi(1 + \sin t) \sqrt{\underbrace{(-\sin t)^2 + (\cos t)^2}_{=1}} dt && \text{about the } x\text{-axis;} \\ &= 2\pi \int_0^{2\pi} (1 + \sin t) dt && y = 1 + \sin t > 0 \\ &= 2\pi[t - \cos t]_0^{2\pi} = 4\pi^2. \end{aligned}$$

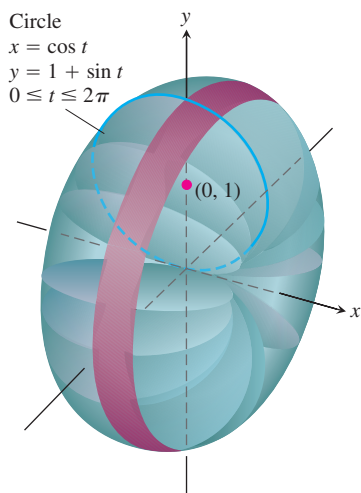


FIGURE 6.50 In Example 3 we calculate the area of the surface of revolution swept out by this parametrized curve.

The Differential Form

The equations

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{and} \quad S = \int_c^d 2\pi x \sqrt{\left(\frac{dx}{dy}\right)^2} dy$$

are often written in terms of the arc length differential $ds = \sqrt{dx^2 + dy^2}$ as

$$S = \int_a^b 2\pi y ds \quad \text{and} \quad S = \int_c^d 2\pi x ds.$$

In the first of these, y is the distance from the x -axis to an element of arc length ds . In the second, x is the distance from the y -axis to an element of arc length ds . Both integrals have the form

$$S = \int 2\pi(\text{radius})(\text{band width}) = \int 2\pi\rho ds \quad (7)$$

where ρ is the radius from the axis of revolution to an element of arc length ds (Figure 6.51).

In any particular problem, you would then express the radius function ρ and the arc length differential ds in terms of a common variable and supply limits of integration for that variable.

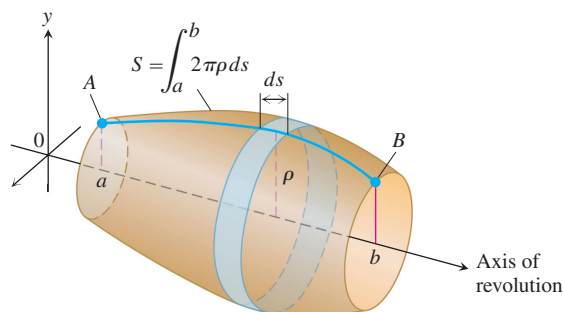


FIGURE 6.51 The area of the surface swept out by revolving arc AB about the axis shown here is $\int_a^b 2\pi\rho ds$. The exact expression depends on the formulas for ρ and ds .

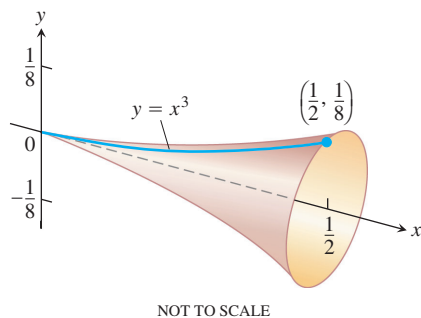


FIGURE 6.52 The surface generated by revolving the curve $y = x^3$, $0 \leq x \leq 1/2$, about the x -axis could be the design for a champagne glass (Example 4).

EXAMPLE 4 Using the Differential Form for Surface Areas

Find the area of the surface generated by revolving the curve $y = x^3$, $0 \leq x \leq 1/2$, about the x -axis (Figure 6.52).

Solution We start with the short differential form:

$$\begin{aligned} S &= \int 2\pi\rho ds \\ &= \int 2\pi y ds \\ &= \int 2\pi y \sqrt{dx^2 + dy^2}. \end{aligned}$$

For revolution about the x -axis, the radius function is $\rho = y > 0$ on $0 \leq x \leq 1/2$.

$$ds = \sqrt{dx^2 + dy^2}$$

We then decide whether to express dy in terms of dx or dx in terms of dy . The original form of the equation, $y = x^3$, makes it easier to express dy in terms of dx , so we continue the calculation with

$$y = x^3, \quad dy = 3x^2 dx, \quad \text{and} \quad \sqrt{dx^2 + dy^2} = \sqrt{dx^2 + (3x^2 dx)^2} \\ = \sqrt{1 + 9x^4} dx.$$

With these substitutions, x becomes the variable of integration and

$$\begin{aligned} S &= \int_{x=0}^{x=1/2} 2\pi y \sqrt{dx^2 + dy^2} \\ &= \int_0^{1/2} 2\pi x^3 \sqrt{1 + 9x^4} dx \\ &= 2\pi \left(\frac{1}{36} \right) \left(\frac{2}{3} \right) (1 + 9x^4)^{3/2} \Big|_0^{1/2} \\ &= \frac{\pi}{27} \left[\left(1 + \frac{9}{16} \right)^{3/2} - 1 \right] \\ &= \frac{\pi}{27} \left[\left(\frac{25}{16} \right)^{3/2} - 1 \right] = \frac{\pi}{27} \left(\frac{125}{64} - 1 \right) \\ &= \frac{61\pi}{1728}. \end{aligned}$$

Substitute
 $u = 1 + 9x^4$,
 $du/36 = x^3 dx$;
integrate, and
substitute back.

Cylindrical Versus Conical Bands

Why not find the surface area by approximating with cylindrical bands instead of conical bands, as suggested in Figure 6.53? The Riemann sums we get this way converge just as nicely as the ones based on conical bands, and the resulting integral is simpler. For revolution about the x -axis in this case, the radius in Equation (7) is $\rho = y$ and the band width is $ds = dx$. This leads to the integral formula

$$S = \int_a^b 2\pi f(x) dx \quad (8)$$

rather than the defining Equation (3). The problem with this new formula is that it fails to give results consistent with the surface area formulas from classical geometry, and that was one of our stated goals at the outset. Just because we end up with a nice-looking integral from a Riemann sum derivation does not mean it will calculate what we intend. (See Exercise 40.)

CAUTION Do not use Equation (8) to calculate surface area. It does *not* give the correct result.

The Theorems of Pappus

In the third century, an Alexandrian Greek named Pappus discovered two formulas that relate centroids to surfaces and solids of revolution. The formulas provide shortcuts to a number of otherwise lengthy calculations.

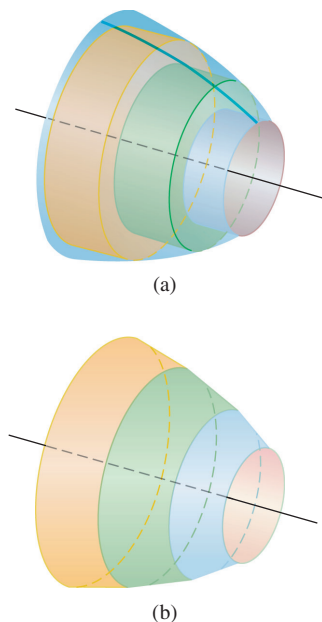


FIGURE 6.53 Why not use (a) cylindrical bands instead of (b) conical bands to approximate surface area?

THEOREM 1 Pappus's Theorem for Volumes

If a plane region is revolved once about a line in the plane that does not cut through the region's interior, then the volume of the solid it generates is equal to the region's area times the distance traveled by the region's centroid during the revolution. If ρ is the distance from the axis of revolution to the centroid, then

$$V = 2\pi\rho A. \quad (9)$$

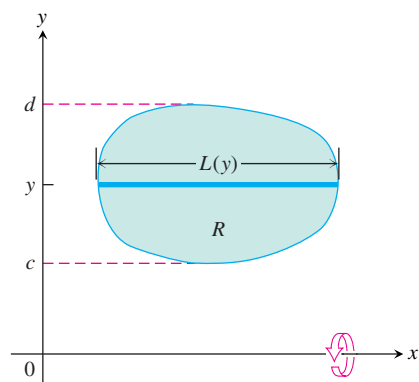


FIGURE 6.54 The region R is to be revolved (once) about the x -axis to generate a solid. A 1700-year-old theorem says that the solid's volume can be calculated by multiplying the region's area by the distance traveled by its centroid during the revolution.

Proof We draw the axis of revolution as the x -axis with the region R in the first quadrant (Figure 6.54). We let $L(y)$ denote the length of the cross-section of R perpendicular to the y -axis at y . We assume $L(y)$ to be continuous.

By the method of cylindrical shells, the volume of the solid generated by revolving the region about the x -axis is

$$V = \int_c^d 2\pi(\text{shell radius})(\text{shell height}) dy = 2\pi \int_c^d y L(y) dy. \quad (10)$$

The y -coordinate of R 's centroid is

$$\bar{y} = \frac{\int_c^d \tilde{y} dA}{A} = \frac{\int_c^d y L(y) dy}{A}, \quad \tilde{y} = y, dA = L(y)dy$$

so that

$$\int_c^d y L(y) dy = A\bar{y}.$$

Substituting $A\bar{y}$ for the last integral in Equation (10) gives $V = 2\pi\bar{y}A$. With ρ equal to $\bar{y}$, we have $V = 2\pi\rho A$. ■

EXAMPLE 5 Volume of a Torus

The volume of the torus (doughnut) generated by revolving a circular disk of radius a about an axis in its plane at a distance $b \geq a$ from its center (Figure 6.55) is

$$V = 2\pi(b)(\pi a^2) = 2\pi^2 b a^2. \quad \blacksquare$$

EXAMPLE 6 Locate the Centroid of a Semicircular Region

Solution We model the region as the region between the semicircle $y = \sqrt{a^2 - x^2}$ (Figure 6.56) and the x -axis and imagine revolving the region about the x -axis to generate a solid sphere. By symmetry, the x -coordinate of the centroid is $\bar{x} = 0$. With $\bar{y} = \rho$ in Equation (9), we have

$$\bar{y} = \frac{V}{2\pi A} = \frac{(4/3)\pi a^3}{2\pi(1/2)\pi a^2} = \frac{4}{3\pi} a. \quad \blacksquare$$

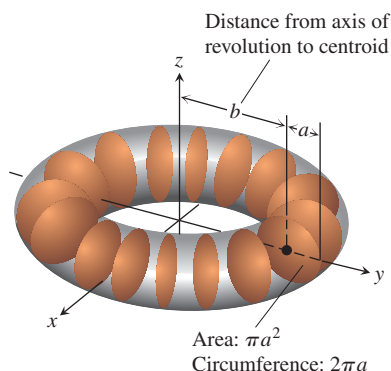


FIGURE 6.55 With Pappus's first theorem, we can find the volume of a torus without having to integrate (Example 5).

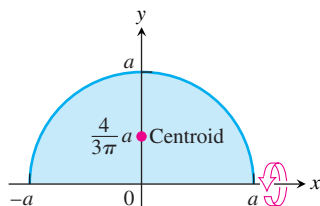


FIGURE 6.56 With Pappus's first theorem, we can locate the centroid of a semicircular region without having to integrate (Example 6).

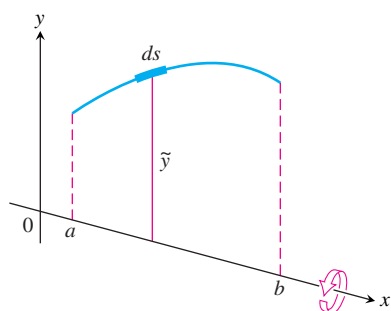


FIGURE 6.57 Figure for proving Pappus's area theorem.

THEOREM 2 Pappus's Theorem for Surface Areas

If an arc of a smooth plane curve is revolved once about a line in the plane that does not cut through the arc's interior, then the area of the surface generated by the arc equals the length of the arc times the distance traveled by the arc's centroid during the revolution. If ρ is the distance from the axis of revolution to the centroid, then

$$S = 2\pi\rho L. \quad (11)$$

The proof we give assumes that we can model the axis of revolution as the x -axis and the arc as the graph of a continuously differentiable function of x .

Proof We draw the axis of revolution as the x -axis with the arc extending from $x = a$ to $x = b$ in the first quadrant (Figure 6.57). The area of the surface generated by the arc is

$$S = \int_{x=a}^{x=b} 2\pi y \, ds = 2\pi \int_{x=a}^{x=b} y \, ds. \quad (12)$$

The y -coordinate of the arc's centroid is

$$\bar{y} = \frac{\int_{x=a}^{x=b} \tilde{y} \, ds}{\int_{x=a}^{x=b} ds} = \frac{\int_{x=a}^{x=b} y \, ds}{L}. \quad \text{L = } \int ds \text{ is the arc's length and } \tilde{y} = y.$$

Hence

$$\int_{x=a}^{x=b} y \, ds = \bar{y}L.$$

Substituting $\bar{y}L$ for the last integral in Equation (12) gives $S = 2\pi\bar{y}L$. With ρ equal to $\bar{y}$, we have $S = 2\pi\rho L$. ■

EXAMPLE 7 Surface Area of a Torus

The surface area of the torus in Example 5 is

$$S = 2\pi(b)(2\pi a) = 4\pi^2 ba. \quad \blacksquare$$

EXERCISES 6.5

Finding Integrals for Surface Area

In Exercises 1–8:

- a. Set up an integral for the area of the surface generated by revolving the given curve about the indicated axis.
 - T** b. Graph the curve to see what it looks like. If you can, graph the surface, too.
 - T** c. Use your grapher's or computer's integral evaluator to find the surface's area numerically.
1. $y = \tan x, \quad 0 \leq x \leq \pi/4; \quad x\text{-axis}$
 2. $y = x^2, \quad 0 \leq x \leq 2; \quad x\text{-axis}$
 3. $xy = 1, \quad 1 \leq y \leq 2; \quad y\text{-axis}$
 4. $x = \sin y, \quad 0 \leq y \leq \pi; \quad y\text{-axis}$
 5. $x^{1/2} + y^{1/2} = 3$ from $(4, 1)$ to $(1, 4); \quad x\text{-axis}$
 6. $y + 2\sqrt{y} = x, \quad 1 \leq y \leq 2; \quad y\text{-axis}$
 7. $x = \int_0^y \tan t \, dt, \quad 0 \leq y \leq \pi/3; \quad y\text{-axis}$
 8. $y = \int_1^x \sqrt{t^2 - 1} \, dt, \quad 1 \leq x \leq \sqrt{5}; \quad x\text{-axis}$

Finding Surface Areas

9. Find the lateral (side) surface area of the cone generated by revolving the line segment $y = x/2$, $0 \leq x \leq 4$, about the x -axis. Check your answer with the geometry formula

$$\text{Lateral surface area} = \frac{1}{2} \times \text{base circumference} \times \text{slant height}.$$

10. Find the lateral surface area of the cone generated by revolving the line segment $y = x/2$, $0 \leq x \leq 4$ about the y -axis. Check your answer with the geometry formula

$$\text{Lateral surface area} = \frac{1}{2} \times \text{base circumference} \times \text{slant height}.$$

11. Find the surface area of the cone frustum generated by revolving the line segment $y = (x/2) + (1/2)$, $1 \leq x \leq 3$, about the x -axis. Check your result with the geometry formula

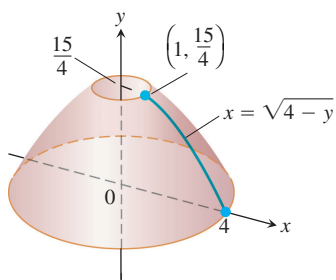
$$\text{Frustum surface area} = \pi(r_1 + r_2) \times \text{slant height}.$$

12. Find the surface area of the cone frustum generated by revolving the line segment $y = (x/2) + (1/2)$, $1 \leq x \leq 3$, about the y -axis. Check your result with the geometry formula

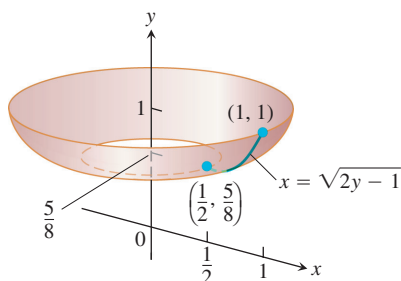
$$\text{Frustum surface area} = \pi(r_1 + r_2) \times \text{slant height}.$$

Find the areas of the surfaces generated by revolving the curves in Exercises 13–22 about the indicated axes. If you have a grapher, you may want to graph these curves to see what they look like.

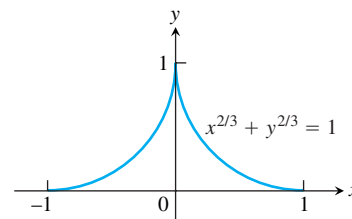
13. $y = x^3/9$, $0 \leq x \leq 2$; x -axis
 14. $y = \sqrt{x}$, $3/4 \leq x \leq 15/4$; x -axis
 15. $y = \sqrt{2x - x^2}$, $0.5 \leq x \leq 1.5$; x -axis
 16. $y = \sqrt{x + 1}$, $1 \leq x \leq 5$; x -axis
 17. $x = y^3/3$, $0 \leq y \leq 1$; y -axis
 18. $x = (1/3)y^{3/2} - y^{1/2}$, $1 \leq y \leq 3$; y -axis
 19. $x = 2\sqrt{4 - y}$, $0 \leq y \leq 15/4$; y -axis



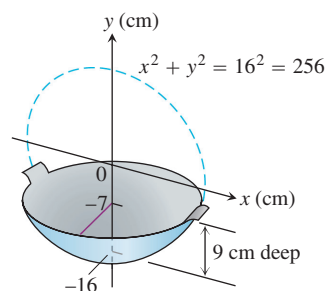
20. $x = \sqrt{2y - 1}$, $5/8 \leq y \leq 1$; y -axis



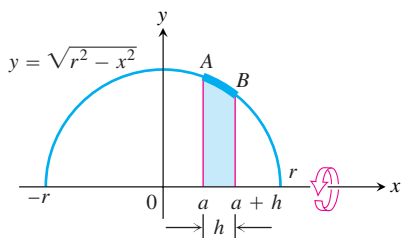
21. $x = (y^4/4) + 1/(8y^2)$, $1 \leq y \leq 2$; x -axis (Hint: Express $ds = \sqrt{dx^2 + dy^2}$ in terms of dy , and evaluate the integral $S = \int 2\pi y ds$ with appropriate limits.)
 22. $y = (1/3)(x^2 + 2)^{3/2}$, $0 \leq x \leq \sqrt{2}$; y -axis (Hint: Express $ds = \sqrt{dx^2 + dy^2}$ in terms of dx , and evaluate the integral $S = \int 2\pi x ds$ with appropriate limits.)
 23. **Testing the new definition** Show that the surface area of a sphere of radius a is still $4\pi a^2$ by using Equation (3) to find the area of the surface generated by revolving the curve $y = \sqrt{a^2 - x^2}$, $-a \leq x \leq a$, about the x -axis.
 24. **Testing the new definition** The lateral (side) surface area of a cone of height h and base radius r should be $\pi r \sqrt{r^2 + h^2}$, the semiperimeter of the base times the slant height. Show that this is still the case by finding the area of the surface generated by revolving the line segment $y = (r/h)x$, $0 \leq x \leq h$, about the x -axis.
 25. Write an integral for the area of the surface generated by revolving the curve $y = \cos x$, $-\pi/2 \leq x \leq \pi/2$, about the x -axis. In Section 8.5 we will see how to evaluate such integrals.
 26. **The surface of an astroid** Find the area of the surface generated by revolving about the x -axis the portion of the astroid $x^{2/3} + y^{2/3} = 1$ shown here. (Hint: Revolve the first-quadrant portion $y = (1 - x^{2/3})^{3/2}$, $0 \leq x \leq 1$, about the x -axis and double your result.)



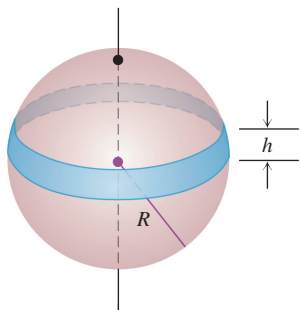
- T 27. Enameling woks** Your company decided to put out a deluxe version of the successful wok you designed in Section 6.1, Exercise 55. The plan is to coat it inside with white enamel and outside with blue enamel. Each enamel will be sprayed on 0.5 mm thick before baking. (See diagram here.) Your manufacturing department wants to know how much enamel to have on hand for a production run of 5000 woks. What do you tell them? (Neglect waste and unused material and give your answer in liters. Remember that $1 \text{ cm}^3 = 1 \text{ mL}$, so $1 \text{ L} = 1000 \text{ cm}^3$.)



- 28. Slicing bread** Did you know that if you cut a spherical loaf of bread into slices of equal width, each slice will have the same amount of crust? To see why, suppose the semicircle $y = \sqrt{r^2 - x^2}$ shown here is revolved about the x -axis to generate a sphere. Let AB be an arc of the semicircle that lies above an interval of length h on the x -axis. Show that the area swept out by AB does not depend on the location of the interval. (It does depend on the length of the interval.)



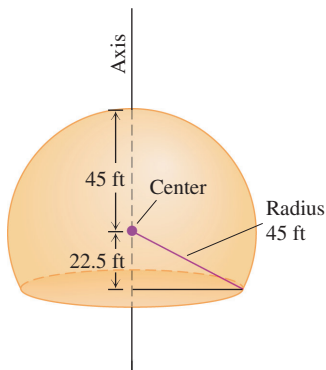
- 29.** The shaded band shown here is cut from a sphere of radius R by parallel planes h units apart. Show that the surface area of the band is $2\pi Rh$.



- 30.** Here is a schematic drawing of the 90-ft dome used by the U.S. National Weather Service to house radar in Bozeman, Montana.

- a. How much outside surface is there to paint (not counting the bottom)?

- T** b. Express the answer to the nearest square foot.



- 31. Surfaces generated by curves that cross the axis of revolution** The surface area formula in Equation (3) was developed under the assumption that the function f whose graph generated the surface was nonnegative over the interval $[a, b]$. For curves that cross the axis of

revolution, we replace Equation (3) with the absolute value formula

$$S = \int 2\pi \rho \, ds = \int 2\pi |f(x)| \, ds. \quad (13)$$

Use Equation (13) to find the surface area of the double cone generated by revolving the line segment $y = x$, $-1 \leq x \leq 2$, about the x -axis.

- 32.** (Continuation of Exercise 31.) Find the area of the surface generated by revolving the curve $y = x^3/9$, $-\sqrt{3} \leq x \leq \sqrt{3}$, about the x -axis. What do you think will happen if you drop the absolute value bars from Equation (13) and attempt to find the surface area with the formula $S = \int 2\pi f(x) \, ds$ instead? Try it.

Parametrizations

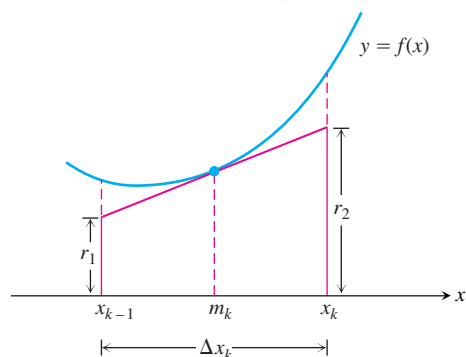
Find the areas of the surfaces generated by revolving the curves in Exercises 33–35 about the indicated axes.

- 33.** $x = \cos t$, $y = 2 + \sin t$, $0 \leq t \leq 2\pi$; x -axis
34. $x = (2/3)t^{3/2}$, $y = 2\sqrt{t}$, $0 \leq t \leq \sqrt{3}$; y -axis
35. $x = t + \sqrt{2}$, $y = (t^2/2) + \sqrt{2}t$, $-\sqrt{2} \leq t \leq \sqrt{2}$; y -axis
36. Set up, but do not evaluate, an integral that represents the area of the surface obtained by rotating the curve $x = a(t - \sin t)$, $y = a(1 - \cos t)$, $0 \leq t \leq 2\pi$, about the x -axis.
37. A cone frustum The line segment joining the points $(0, 1)$ and $(2, 2)$ is revolved about the x -axis to generate a frustum of a cone. Find the surface area of the frustum using the parametrization $x = 2t$, $y = t + 1$, $0 \leq t \leq 1$. Check your result with the geometry formula: Area = $\pi(r_1 + r_2)(\text{slant height})$.
38. A cone The line segment joining the origin to the point (h, r) is revolved about the x -axis to generate a cone of height h and base radius r . Find the cone's surface area with the parametric equations $x = ht$, $y = rt$, $0 \leq t \leq 1$. Check your result with the geometry formula: Area = $\pi r(\text{slant height})$.

- 39. An alternative derivation of the surface area formula** Assume f is smooth on $[a, b]$ and partition $[a, b]$ in the usual way. In the k th subinterval $[x_{k-1}, x_k]$ construct the tangent line to the curve at the midpoint $m_k = (x_{k-1} + x_k)/2$, as in the figure here.

- a. Show that $r_1 = f(m_k) - f'(m_k) \frac{\Delta x_k}{2}$ and $r_2 = f(m_k) + f'(m_k) \frac{\Delta x_k}{2}$.

- b. Show that the length L_k of the tangent line segment in the k th subinterval is $L_k = \sqrt{(\Delta x_k)^2 + (f'(m_k) \Delta x_k)^2}$.

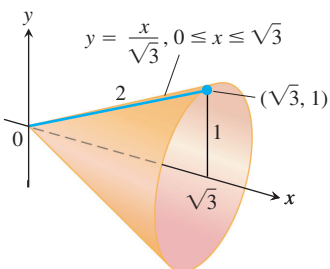


c. Show that the lateral surface area of the frustum of the cone swept out by the tangent line segment as it revolves about the x -axis is $2\pi f(m_k)\sqrt{1 + (f'(m_k))^2} \Delta x_k$.

d. Show that the area of the surface generated by revolving $y = f(x)$ about the x -axis over $[a, b]$ is

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\begin{array}{c} \text{lateral surface area} \\ \text{of } k\text{th frustum} \end{array} \right) = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx.$$

- 40. Modeling surface area** The lateral surface area of the cone swept out by revolving the line segment $y = x/\sqrt{3}$, $0 \leq x \leq \sqrt{3}$, about the x -axis should be $(1/2)(\text{base circumference})(\text{slant height}) = (1/2)(2\pi)(2) = 2\pi$. What do you get if you use Equation (8) with $f(x) = x/\sqrt{3}$?



The Theorems of Pappus

- 41.** The square region with vertices $(0, 2)$, $(2, 0)$, $(4, 2)$, and $(2, 4)$ is revolved about the x -axis to generate a solid. Find the volume and surface area of the solid.
- 42.** Use a theorem of Pappus to find the volume generated by revolving about the line $x = 5$ the triangular region bounded by the coordinate axes and the line $2x + y = 6$. (As you saw in Exercise 29 of Section 6.4, the centroid of a triangle lies at the intersection of

the medians, one-third of the way from the midpoint of each side toward the opposite vertex.)

- 43.** Find the volume of the torus generated by revolving the circle $(x - 2)^2 + y^2 = 1$ about the y -axis.
- 44.** Use the theorems of Pappus to find the lateral surface area and the volume of a right circular cone.
- 45.** Use the Second Theorem of Pappus and the fact that the surface area of a sphere of radius a is $4\pi a^2$ to find the centroid of the semicircle $y = \sqrt{a^2 - x^2}$.
- 46.** As found in Exercise 45, the centroid of the semicircle $y = \sqrt{a^2 - x^2}$ lies at the point $(0, 2a/\pi)$. Find the area of the surface swept out by revolving the semicircle about the line $y = a$.
- 47.** The area of the region R enclosed by the semiellipse $y = (b/a)\sqrt{a^2 - x^2}$ and the x -axis is $(1/2)\pi ab$ and the volume of the ellipsoid generated by revolving R about the x -axis is $(4/3)\pi ab^2$. Find the centroid of R . Notice that the location is independent of a .
- 48.** As found in Example 6, the centroid of the region enclosed by the x -axis and the semicircle $y = \sqrt{a^2 - x^2}$ lies at the point $(0, 4a/3\pi)$. Find the volume of the solid generated by revolving this region about the line $y = -a$.
- 49.** The region of Exercise 48 is revolved about the line $y = x - a$ to generate a solid. Find the volume of the solid.
- 50.** As found in Exercise 45, the centroid of the semicircle $y = \sqrt{a^2 - x^2}$ lies at the point $(0, 2a/\pi)$. Find the area of the surface generated by revolving the semicircle about the line $y = x - a$.
- 51.** Find the moment about the x -axis of the semicircular region in Example 6. If you use results already known, you will not need to integrate.